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Abstract

We present a randomized algorithm for selecting the median of an unsorted array in $\mathcal{O}(n)$ time with high probability. Unlike classical deterministic methods—notably the median-of-medians algorithm—our approach leverages random sampling to simplify the pivot-selection process and reduce overhead. Specifically, we extract a sublinear sample of the input, sort this sample, and employ its near-median elements as pivots. By bounding $\mathbb{P}[|X - \mathbb{E}[X]| \geq k]$ for appropriate choices of X and k, we ensure that the size of the sub-array requiring a full sort remains sufficiently small. A rigorous analysis, based on Chebyshev's inequality, demonstrates that the algorithm's probability of "failure" (i.e., incorrectly bracketed pivots) decays quickly with the input size. In successful executions, the asymptotic cost of sorting the sample and the sub-array is dominated by a single pass through the input, yielding an overall running time of $\mathcal{O}(n)$. While deterministic linear-time solutions to selection exist, our randomized method offers competitive performance in practice, owing to its simplicity and favorable hidden constants. We also discuss how this technique can be combined with deterministic strategies to eliminate worst-case risk.

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Chapter 1

Introduction

1.1 Introduction

1.1.1 Motivation and Related Work

Graph colouring is a fundamental problem in computer science. We are given a graph and we would like to colour the vertices using as few colours overall as possible such that no two adjacent vertices get the same colour. Deciding if a graph can be coloured using $k = 2$ colours is easy, as it is equivalent to deciding if the graph is bipartite. However, for $k \geq 3$, it has been known for a long time that deciding whether k colours suffices is NP-hard **?**.

Contrasting this worst-case hardness result, there has been a lot of work on average-case settings, e.g. **?**. In these settings, the input contains randomness, a colouring is guaranteed to exist, and we aim to recover one with high probability. For the fully random case – when the initial graph is -and a k-colouring is randomly *planted* in it by assigning colours at random and removing edges in-between the colour classes – **?** provided an efficient algorithm.

The key question then is what happens between worst-case and average-case – how much can the full-randomness assumptions of **?** be relaxed? Towards this goal, **?** looked at semi-random models where either the initial graph or the planted colouring is random. Under the crucial assumption that the initial graph is always a two-sided expander – the second largest and smallest eigenvalues of its normalised adjacency matrix are bounded away from 1 in absolute value – the authors provided a coarse-grained characterisation of when there is an efficient algorithm and when it is NP-hard. Then, **?** derandomised this result, pinpointing deterministic pseudorandom conditions.

The algorithms in all the aforementioned papers are based on the observation that there is an encoding of a colouring that is roughly a linear combination of eigenvectors with a very negative eigenvalue. Therefore, when the adjacency matrix has only a few very negative eigenvalues, an approximate colouring can be decoded after subspace enumeration **?**.

The case of *one*-sided expansion – when only the second largest eigenvalue is guaranteed to be bounded away from 1 – is therefore much more challenging, and it is also less studied. In a recent paper, **?** showed that for $k = 3$ specifically, a small linear-sized independent set can be recovered – very far away from a complete colouring. Their algorithm is based on showing within the sum-of-squares proof system **?** that under one-sided expansion – unless the graph is almost-bipartite – any two 3-colourings must be sufficiently similar.

For completeness, note that graph colouring can be studied from other perspectives as well. For example, given a 3-colourable graph, instead of finding assumptions under which a 3-colouring can be recovered, **?** provide a more general algorithm that uses $\tilde{O}(n^{0.19747})$ colours.

1.1.2 Results

In this work, we provide an almost full characterisation of recovering k-colourings in *one*-sided expanders.

Key to our results is the concept of the model matrix of a graph that contains the average degrees between the colour classes. In **??**, we provide an algorithm that given a one-sided expander graph whose model matrix has no repeated rows, outputs an almost complete colouring. The more general **??** recovers a *partial* colouring of size dependent on the relative sizes of the colour classes corresponding to repeated rows in the model matrix.

As corollaries, we show in **??** that the two sides of an *almost* bipartite one-sided expander graph can be almost completely recovered. Then, for $k = 3$ we prove in **??** that when the colour classes are not very unbalanced, an almost complete colouring can be recovered. Then, we show in **??** that for any $k = O(1)$, when a random colouring is planted in a one-sided expander graph, a *complete* colouring can be recovered.

As a side result, we show in **??** that in a one-sided expander that contains an independent set with almost half of the vertices, an independent set in size close to that can be recovered – this improves on the result of **?** that can only recover a small linear-sized independent set.

Finally, in **??** we provide hardness results corresponding to most of our aforementioned results, showing that the assumptions on the model matrix, one-sided expansion, and balancedness are indeed necessary.

1.1.3 Techniques

Let $\lambda_1, \ldots, \lambda_k$ and $v_1, \ldots, v_k$ represent the k eigenvalues and eigenvectors respectively of the model matrix M associated with the colour classes of the k-colourable graph. Define $u_1, \ldots, u_k$ as the corresponding n-dimensional vectors, where vertices sharing the same colour correspond to identical entries in the respective v. Then, the core idea behind **??** – which is the basis of most of our results – is the following:

1. In **??** we show that if all rows of M are (sufficiently) distinct, then for any two indices $1 \leq x < y \leq k$, there is an eigenvector v_i corresponding to a *non-zero* eigenvalue that distinguishes them (i.e. $v_i[x] \neq v_i[y]$). This means that if we can (approximately) recover the vectors u_i corresponding to eigenvectors of M with non-zero eigenvalues, we can (approximately) recover the k-colouring by a simple clustering algorithm.
2. To be able recover the aforementioned vectors u_i, we first show in **??** that they are *almost* eigenvectors of the adjacency matrix A of the graph with eigenvalues λ_i bounded away from zero (i.e. $Au_i \approx \lambda_i u_i$). This result is based on a surprising connection: if the graph is a one-sided expander, then the the variance of the degrees between any two colour classes is small – intuitively, the adjacency matrix is close to a blow-up of the model matrix.
3. We recover these vectors u_i approximately by subspace enumeration in the eigenspace of the adjacency matrix bounded away from zero. For this procedure to be efficient, we need that the adjacency matrix has only few very negative eigenvalues – this is not guaranteed by assumption, as we only make an assumption about the second largest eigenvalue. In **??** we show a surprising more general result that guarantees it nevertheless: for any symmetric matrix with non-negative entries, small top threshold rank implies small bottom threshold rank.

To deduce **??** from **??**, we observe in **??** that for $k = 3$, given the colour class sizes, the model matrix is fixed due to regularity, and unless one colour class contains almost half of the vertices, it will have no repeated rows.

The proof for approximate recovery in the random planted setting (**??**) is similar to the proof of **??**, but note that while the original graph is a one-sided expander, the resulting graph after planting might no longer be expanding. Hence, to show low variance of the degrees between colour classes, instead of **??** we exploit full randomness using concentration bounds. Efficient subspace enumeration is guaranteed by **??** that shows that low threshold rank is preserved after planting. To convert an approximate colouring into a complete colouring (**??**) we adapt **?** by showing that the expander mixing lemma and small-set vertex expansion – on which their algorithms rely – essentially holds for one-sided expansion as well (**??**, **??**).

The proof of **??** relies on the observation that if an independent set is nearly half the size of the graph, regularity ensures most edges connect the independent set to the rest of the graph. Consequently, the ± 1 vector representing this partition is approximately an eigenvector of the adjacency matrix with a very negative eigenvalue.

Finally, the hardness results in **??** build on **??** (from **?**) which shows that without any expansion assumption, it is NP-hard (under the Unique Games Conjecture) to recover even a linear-sized independent set in an almost-bipartite graph.

1.1.4 An introductory example

To familiarise the reader with the topic, we first provide a solution for a simple question that was "left as an exercise to the reader" in **?**:

Lemma 1.1 (Linear degree random planting full recovery) *Let H be an n-vertex graph with minimum degree $\Omega(n)$. Let G be the graph obtained by randomly planting a 3-colouring in H. Then, there exists a polynomial-time algorithm that recovers a complete 3-colouring of G.*

The intuition behind this result is that more edges provides more constraints, enabling the recovery of vertex colours through a process of elimination. Surprisingly, neither of our results in this work will require the degrees to be even $\omega(1)$ – we will, however, require regularity and expansion.

Proof (Proof of ??) First we provide the algorithm, then we show that it works with high probability.

Algorithm 1 Linear Minimum Degree Algorithm

Select $S \subseteq V$ uniformly at random such that $|S| = \Theta(\log n)$. Assert that each vertex $x \notin S$ has at least one neighbour in S.

for each of the $3^{|S|}$ ways of assigning colours to the vertices in S **do**

  for each vertex $y \notin S$ **do** Exclude from y's possible colours any colour assigned to its neighbours in S. Check if the graph is colourable using at most two remaining colours for each vertex via 2-SAT.

Let $d_H(x)$ and $d_G(x)$ be the degrees of vertex x in H and G respectively. As each edge is deleted with probability $1/3$, by linearity of expectation we have $\mathbb{E}[=]\frac{2}{3}d_H(x)$ for any $x \in V(G)$. Then, by the Chernoff bound, since $d_G(x)$ is a sum of $\Omega(n)$ independent Bernoulli random variables, with high probability we get $d_G(x) \geq \frac{d_H(x)}{3} = \Omega(n)$ for all $x \in V(G)$.

Since the algorithm goes through all possible ways of assigning a colour to the vertices in S, at least one of these will align with the planted colouring, i.e., it will be possible to still 3-colour the remaining vertices. It only remains to show that, , the assertion in Line 2 of **??** will hold, so we can indeed use 2-SAT:

$$\begin{aligned}
\P\left[\exists x \mid N_G(x) \cap S = \emptyset\right] &\leq \sum_{x \in V(G)} \left(1 - \frac{d_G(x)}{n}\right)^S && \text{by union bound} \\
&\leq \sum_{x \in V(G)} e^{-\frac{S d_G(x)}{n}} && \text{using } 1 - x \leq e^{-x} \\
&\leq \sum_{x \in V(G)} e^{-2\log n} && \text{since } d_G(x) = \Omega(n) \text{ and } S \geq O(\log n) \\
&= n^{-1}\,.
\end{aligned}$$

The remaining colouring problem, where each vertex has at most two possible colours, can be efficiently solved using 2-SAT. The formulation involves three types of clauses:

- Each vertex x with possible colours a and b must be assigned a colour: $x_a \vee x_b$.
- Each vertex x must be assigned a unique colour: $\neg x_a \vee \neg x_b$.
- Adjacent vertices x and y must not share the same colour c: $\neg x_c \vee \neg y_c$.

Here, x_c is a boolean variable indicating whether vertex x is assigned colour c.

The runtime of the algorithm is polynomial, as the outer for loop goes through $3^{O(\log n)} = n^{O(1)}$ combinations and 2-SAT is solvable in polynomial time.

Chapter 2

Notation and Preliminaries

2.1 Definitions and notation

For a graph G with adjacency matrix A and (diagonal) degree matrix D with $D_{ii} = \deg(i)$, we say it is a λ-one-sided expander if $\lambda_2(D^{-1/2}AD^{-1/2}) \leq \lambda$. We say a matrix $A \in \mathbb{R}_{\geq 0}^{n\times n}$ is *row stochastic* if the sum of its entries on every row is 1: for all $x \in [n]$, $\sum_{y=1}^{n} A_{xy} = 1$. We say a matrix $A \in \mathbb{R}_{\geq 0}^{n\times n}$ is *reversible row stochastic* if it is row stochastic and if there exists $\pi \in \mathbb{R}_{\geq 0}^{n}$ such that, for all $x, y \in [n]$, $\pi_x A_{xy} = \pi_y A_{yx}$.

Our main results will be stated in terms of (k, δ)-colourings.

Definition 2.1 ((k, δ)-colouring) *Let G be an n-vertex graph. A (k, δ)-colouring of G is a list of k disjoint independent sets in G that cover $(1 - \delta)n$ of the vertices of G.*

We define now the partition matrices and the model of a graph, which are defined with respect to a k-partition $\chi : [n] \to [k]$ of its vertices.

Definition 2.2 (Partition matrices of a colouring) *Let $\chi : [n] \to [k]$ be a k-partition of $[n]$. Then we define the following matrices:*

- *The partition matrix $Z \in 0, 1^{n\times k}$ has $Z_{x\chi(x)} = 1$ for all $x \in [n]$ and 0 elsewhere.*
- *The partition size matrix $B \in \mathbb{R}^{k\times k}$ is the diagonal matrix $B = Z^\top Z$. Note that $B_{aa} = |\chi^{-1}(a)|$ for all $a \in [k]$.*

Definition 2.3 (Model of a colouring) *Let $A \in \mathbb{R}^{n\times n}$, and let $\chi : [n] \to [k]$ be a k-partition of $[n]$. For $x \in [n]$ and $a \in [k]$, let $xa = \sum_{y\in\chi^{-1}(a)} A_{xy}$. We define the* model *$M(A, \chi) \in \mathbb{R}^{k\times k}$ to be the matrix that, for any $a, b \in [k]$, has entries $M(A, \chi)_{ab} = \frac{1}{|\chi^{-1}(a)|} \sum_{x\in\chi^{-1}(a)} xb$.*

Remark 2.4 *Using the definitions in* **??**, *we have equivalently that* $M(A, \chi) = B^{-1}Z^\top AZ$.

For a reversible row stochastic matrix $M \in \mathbb{R}^{k\times k}_{\geq 0}$, we define an (M, δ)-colouring, which is a colouring with model close to M:

Definition 2.5 ((M, δ)-colourable graph) *Let $M \in \mathbb{R}^{k\times k}_{\geq 0}$ be a reversible row stochastic matrix with zeros on the diagonal. Let G be an n-vertex graph with adjacency matrix A and (diagonal) degree matrix D with $D_{ii} = \deg(i)$, and let $\tilde{A} = D^{-1/2}AD^{-1/2}$. We say that G is (M, δ)-colourable if there exists a k-partition $\chi : [n] \to [k]$ such that $M - M(\tilde{A}, \chi)_{\max} \leq \delta$ and G has a vertex cover of all edges with both endpoints in the same part of size at most δn with respect to χ.*

Chapter 3

Spectral Result

3.1 Small top threshold rank implies small bottom threshold rank

In this section, we prove the following result which shows that, for bounded-norm symmetric matrices with non-negative entries, large bottom threshold rank implies large top threshold rank. In the rest of the paper, we will be applying this result to normalized adjacency matrices of graphs, which indeed are symmetric, have spectral norm 1, and have non-negative entries.

Theorem 3.1 (Large bottom rank implies large top rank) *Let $A \in \mathbb{R}_{\geq 0}^{n\times n}$ be symmetric with $A \leq 1$. If*

$$_{\leq -\lambda}(A) \geq t\,,$$

for some $\lambda \geq 0$ and some positive integer t, then for any $\sigma \in (0,1)$ it follows that

$$_{\geq \frac{\lambda^2-\sigma}{1-\sigma}}(A) \geq \sigma^2 t\,.$$

By taking the contrapositive of **??** and re-parameterizing, we get the following direct corollary that shows, for bounded-norm symmetric matrices with non-negative entries and small top threshold rank, that the bottom threshold rank cannot be too large.

Corollary 3.2 (Small top rank implies small bottom rank) *Let $A \in \mathbb{R}_{\geq 0}^{n\times n}$ be symmetric with $A \leq 1$. If*

$$_{\geq \tau}(A) \leq s\,,$$

for some $\tau \geq 0$ and some non-negative integer s, then for any $\sigma \in (0,1)$ it follows that

$$_{\leq -\sqrt{\tau(1-\sigma)+\sigma}}(A) \leq \frac{s}{\sigma^2}\,.$$

To prove **??**, we need the following slight generalization of Lemma 6.1 in **?**. We include a proof of this lemma in **??** for completeness.

Lemma 3.3 (Generalized restatement of Lemma 6.1 in ?) *Let $A \in \mathbb{R}^{n\times n}$ be symmetric with $A \leq 1$. Suppose there exists $M \in \mathbb{R}^{n\times n}$ positive semidefinite such that*

$$A, M \geq 1-, \quad M_F^2 \leq \frac{1}{r}, \quad (M) = 1 .$$

Then for all $C > 1$,

$${}_{\geq 1-C\cdot}(A) \geq (1 - \frac{1}{C})^2 r .$$

The key observation that leads to the proof of **??** is that, if a bounded-norm symmetric matrix with non-negative entries has large bottom threshold rank, then we can construct a matrix M that satisfies **??**. This idea is captured by the following lemma.

Lemma 3.4 (Construction of witness matrix) *Let $A \in \mathbb{R}^{n\times n}_{\geq 0}$ be symmetric with $A \leq 1$. If*

$${}_{\leq -\lambda}(A) \geq t ,$$

for some $\lambda \geq 0$ and some positive integer t, then there exist $V \in \mathbb{R}^{t^2\times n}$ such that

$$A, V^\top V \geq \lambda^2 , \quad V^\top V_F^2 \leq \frac{1}{t} , \quad (V^\top V) = 1 .$$

Proof Let $u_1, u_2, \ldots, u_t$ be t orthonormal eigenvectors of A whose corresponding eigenvalues are at most $-\lambda$, that is, for each $s \in [t]$,

$$u_s, Au_s \leq -\lambda \quad \text{and} \quad \|u_s\|^2 = 1 .$$

Let $U \in \mathbb{R}^{n\times t}$ be the matrix whose s-th column is $\frac{1}{\sqrt{t}} u_s$ for all $s \in [t]$. It follows that

$$A, UU^\top = \sum_{s\in[t]} \frac{1}{t} A, u_s(u_s)^\top = \frac{1}{t} \sum_{s\in[t]} u_s, Au_s \leq -\lambda , \tag{3.1}$$

and

$$UU^{\top}{}_F^2 = U^\top U_F^2 = \sum_{s\in[t]} \frac{1}{t^2} \|u_s\|^4 = \frac{1}{t} , \tag{3.2}$$

and

$$(UU^\top) = (U^\top U) = \sum_{s\in[t]} \frac{1}{t} \|u_s\|^2 = 1 . \tag{3.3}$$

Let us denote the i-th row of U by w_i. For $i \in [n]$, define $v_i = \frac{w_i^{\otimes 2}}{\|w_i\|}$ if $\|w_i\| \neq 0$ and $v_i = 0$ (the all-zero vector) otherwise. Let $S = i \in [n] : \|w_i\| \neq 0$. Finally, let $V \in \mathbb{R}^{t^2\times n}$ be the matrix whose i-th column is v_i for all $i \in [n]$. We will show that V satisfies the three conditions of the lemma.

Condition 1. Notice that, by Cauchy-Schwarz, we have

$$\begin{aligned}(\sum_{i,j\in S} A_{ij}w_i,w_j)^2 &= (\sum_{i,j\in S} \frac{\sqrt{A_{ij}}\cdot w_i,w_j}{\sqrt{\|w_i\|\|w_j\|}}\cdot\sqrt{A_{ij}\|w_i\|\|w_j\|})^2\\ &\le (\sum_{i,j\in S} A_{ij}\frac{w_i,w_j^2}{\|w_i\|\|w_j\|})\cdot(\sum_{i,j\in S} A_{ij}\|w_i\|\|w_j\|)\,.\end{aligned}$$

By rearranging, we get

$$A,V^\top V = \sum_{i,j} A_{ij}v_i,v_j = \sum_{i,j\in S} A_{ij}\frac{w_i,w_j^2}{\|w_i\|\|w_j\|} \ge \frac{(\sum_{i,j\in S} A_{ij}w_i,w_j)^2}{\sum_{i,j\in S} A_{ij}\|w_i\|\|w_j\|}\,. \tag{3.4}$$

For the numerator, by observing that $w_i,w_j = (UU^\top)_{ij}$ and by **??**, we get

$$(\sum_{i,j\in S} A_{ij}w_i,w_j)^2 = A,UU^{\top 2} \ge \lambda^2\,. \tag{3.5}$$

For the denominator, because $A \le 1$ we get

$$\sum_{i,j\in S} A_{ij}\|w_i\|\|w_j\| \le \sum_{i\in S}\|w_i\|^2 = (UU^\top) = 1\,. \tag{3.6}$$

Plugging **??** and **??** into **??**, we get that V satisfies condition 1,

$$A,V^\top V \ge \lambda^2\,.$$

Condition 2. By the definition of V and by Cauchy-Schwarz, we get

$$V^\top V_F^2 = \sum_{i,j} v_i,v_j^2 = \sum_{i,j\in S}\frac{w_i,w_j^4}{\|w_i\|^2\|w_j\|^2} \le \sum_{i,j\in S} w_i,w_j^2\,.$$

Since $w_i,w_j = (UU^\top)_{ij}$, it follows by **??** that

$$V^\top V_F^2 \le \sum_{i,j\in S} w_i,w_j^2 = UU^\top{}_F^2 = \frac{1}{t}\,.$$

Condition 3. By the definition of V and **??**, it follows that

$$(V^\top V) = \sum_i \|v_i\|^2 = \sum_i \|w_i\|^2 = (UU^\top) = 1\,.$$

Given **??** and **??**, we can prove **??** easily.

Proof (Proof of ??) By **??**, when ${}_{\le -\lambda}(A) \ge t$, there exist $V \in \mathbb{R}^{t^2\times n}$ such that

$$A,V^\top V \ge \lambda^2\,,\quad V^\top V_F^2 \le \frac{1}{t}\,,\quad (V^\top V) = 1\,.$$

Then, we can apply **??** by setting $M = V^\top V$, $= 1-\lambda^2$, $r = t$, and $C = \frac{1}{1-\sigma}$, and obtain

$${}_{\ge \frac{\lambda^2-\sigma}{1-\sigma}}(A) \ge \sigma^2 t\,.$$

Chapter 4

k-Partitioning

4.1 Recovering partitions

In this section we state and prove our meta-theorem for recovering partitions. The result in this section is written in a very general form, to make clear what properties we exploit in the proof. Then, in **??**, we apply this meta-theorem to our settings of interest and obtain our algorithmic results.

Theorem 4.1 (Result for recovering partitions) *Let $\chi : [n] \to [k]$ be a k-partition of $[n]$, and consider its partition matrices $Z \in \{0,1\}^{n\times k}$ and $B \in \mathbb{R}^{k\times k}$ as per* **??**. *Suppose $|\chi^{-1}(a)| \geq cn$ for all $a \in [k]$. Let $M \in \mathbb{R}^{k\times k}$ be a matrix such that*

- *MB^{-1} is symmetric,*
- *$M \leq \zeta$,*
- *$M^{(a)} - M^{(b)} \geq \alpha$ for any two rows $M^{(a)}$ and $M^{(b)}$ of M.*

Also let $W = ZMB^{-1}Z^{\top}$ and let $A \in \mathbb{R}^{n\times n}$ be some matrix such that, for all $a \in [k]$,

$$A\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq \delta\,,$$

and suppose that $_{\leq -\lambda}(A) +_{\geq\lambda} (A) \leq t$ for some $\lambda > 0$ such that $\lambda \leq \alpha\sqrt{c}/4$ and $\lambda \leq \zeta/4$.

Then, given A, there exists an algorithm that runs in time $n^{O(1)} \cdot (O_{\alpha,\delta,\zeta,c,k}(1))^t$ and outputs a list of $(O_{\alpha,\delta,\zeta,c,k,t}(1))^t$ k-partitions $\hat{\chi} : [n] \to [k]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim \mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O(\frac{\delta\zeta^2 k^3}{\min\{\alpha^2 c,\zeta^2\}\alpha^2 c})$.

In our applications, we will take M to be a model of the colouring χ, as per **??**. Our main results for graph colouring, in particular, take $M = M(\tilde{A}, \chi)$, where $\tilde{A}$ is the normalized adjacency matrix of the graph. Then M_{ab} corresponds to the transition probability from colour class a to colour class b. (An exception is our result for randomly planted graph colourings in **??**, where for technical reasons we take $M = \frac{1}{d'}A$, where A is the adjacency matrix of the graph and d' is its *expected* average degree.) We note that $W_{ij} = M_{\chi(i)\chi(j)}/|\chi^{-1}(\chi(j))|$ is simply the extension of M to the entire matrix, according to the colour classes, up to the normalization factor $1/|\chi^{-1}(\chi(j))|$.

The proof of **??** is a straightforward combination of the following three independent lemmas.

Lemma 4.2 (Eigenspace approximation) *Let $\chi : [n] \to [k]$ be a k-partition of $[n]$, and consider its partition matrix $Z \in \{0,1\}^{n\times k}$ as per* **??***. Let $W \in \mathbb{R}^{n\times n}$ be a symmetric matrix such that W has column span in the span of $Z_{a a\in[k]}$. Alo let $A \in \mathbb{R}^{n\times n}$ be some matrix such that, for all $a \in [k]$,*

$$A\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq \delta\,,$$

and suppose that $_{\leq -\lambda}(A) +_{\geq\lambda}(A) \leq t$ for some $\lambda > 0$. Then, given A, for any $\eta > 0$, there exists an algorithm that runs in time $n^{O(1)} \cdot (\delta k/\eta^2)^{-O(t)}$ and outputs a list of size $(\delta k/\eta^2)^{-O(t)}$ of unit vectors such that, for each unit vector $u \in \mathbb{R}^n$ in the eigenspace of W corresponding to eigenvalues larger than $\lambda + \eta$ in absolute value, there exists some $\hat{u} \in \mathbb{R}^n$ in the list such that $\hat{u} - u^2 \leq O(\delta k/\eta^2)$.

Lemma 4.3 (Eigenvector coordinate separation) *Let $\chi : [n] \to [k]$ be a k-partition of $[n]$, and consider its partition matrices $Z \in \{0,1\}^{n\times k}$ and $B \in \mathbb{R}^{k\times k}$ as per* **??***. Suppose $|\chi^{-1}(a)| \geq cn$ for all $a \in [k]$. Let $M \in \mathbb{R}^{k\times k}$ be a matrix such that*

- *MB^{-1} is symmetric,*
- *$M \leq \zeta$,*
- *$M^{(a)} - M^{(b)} \geq \alpha$ for any two rows $M^{(a)}$ and $M^{(b)}$ of M.*

For some $\lambda \leq \alpha\sqrt{c}/2$ and $\lambda \leq \zeta/2$, let the unit vectors $v_1, \ldots, v_r \in \mathbb{R}^k$ be the eigenvectors of M corresponding to eigenvalues larger than λ in absolute value. Then

$$\min_{a\neq b\in[k]} \max_{i\in[r]} \frac{1}{Zv_i}(v_i)_a - (v_i)_b \geq \Omega(\frac{\alpha\sqrt{c}}{\zeta\sqrt{kn}})\,.$$

Lemma 4.4 (Clustering with separated rows) *Let $\chi : [n] \to [k]$ be a k-partition of $[n]$, and consider its partition matrix $Z \in \{0,1\}^{n\times k}$ as per* **??***. Suppose $|\chi^{-1}(a)| \geq cn$ for all $a \in [k]$. Let unit vectors $v_1, \ldots, v_r \in \mathbb{R}^k$ satisfy*

$$\min_{a\neq b\in[k]} \max_{i\in[r]} \frac{1}{Zv_i}(v_i)_a - (v_i)_b \geq \frac{\alpha}{\sqrt{n}}\,.$$

Then, given unit vectors $\hat{u}_1, \ldots, \hat{u}_r \in \mathbb{R}^n$ such that $\hat{u}_i - \frac{Zv_i}{Zv_i}^2 \leq \delta$ for all $i \in [r]$, there exists an algorithm that runs in time $n^{O(1)} \cdot (\alpha\sqrt{c})^{-O(k^2)}$ and outputs a list of

$(\alpha\sqrt{c})^{-O(k^2)}$ k-partitions $\hat{\chi} : [n] \to [k]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim\mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O(\frac{\delta k}{\alpha^2})$.

Let us now prove **??** given **??**, **??**, and **??**.

Proof (Proof of ??) We will assume $\lambda = \min\{\alpha\sqrt{c}/4, \zeta/4\}$, such that 2λ satisfies the upper bound required to apply **??**.

First, let $S := u \in \mathbb{R}^n \mid Wu = \tau u, u = 1, \tau > 2\lambda$ be the set of eigenvectors of W with corresponding eigenvalues larger than 2λ in absolute value. By applying **??** with $\eta = \lambda$, we can recover a list of size $(\delta k/\lambda^2)^{-O(t)}$ such that, for each unit eigenvector $u \in \mathbb{R}^n$ in this subspace, there exists some $\hat{u}$ in the list with $\|\hat{u} - u\|^2 \leq O(\delta k/\lambda^2)$.

Observe that each eigenvector u of W is by definition in the column span of Z, so we can write $u = Zv$ for some $v \in \mathbb{R}^k$. Suppose $Wu = \tau u$. Then $Mv = B^{-1}Z^\top WZv = \tau B^{-1}Z^\top Zv = \tau v$, so v is an eigenvector of M with eigenvalue τ. Hence, if $v_1, \dots, v_r$ are the eigenvalues of M with corresponding eigenvalues larger than 2λ in absolute value, then $\frac{Zv_1}{Zv_1}, \dots, \frac{Zv_r}{Zv_r} \in S$.

By **??**, we have for these eigenvectors $v_1, \dots, v_r$ of M that

$$\min_{a\neq b\in[k]} \max_{i\in[r]} \frac{1}{Zv_i}(v_i)_a - (v_i)_b \geq \Omega(\frac{\alpha\sqrt{c}}{\zeta\sqrt{kn}}),$$

and by the previous discussion for each v_i there exists some $\hat{u} \in \mathbb{R}^n$ in the recovered list such that $\hat{u} - \frac{Zv_i}{Zv_i}^2 \leq O(\delta k/\lambda^2)$. To know which of the unit vectors $\hat{u}$ in the list correspond to these eigenvectors, we can afford to try all possibilities of at most k vectors in the list; there are at most $(\delta k/\lambda^2)^{-O(tk)}$ such possibilities. The guarantee then follows from **??**, giving probability of misclassification $O(\frac{\delta\zeta^2 k^3}{\min\{\alpha^2 c, \zeta^2\}\alpha^2 c})$. □

The next three subsection prove **??**, **??**, and **??**.

4.1.1 Eigenspace approximation

In this section we prove **??**. First, we state a generic result that proves that if two matrices are close in a subspace that for one of the matrices is an eigenspace with eigenvalues of large absolute value, then the small eigenvalues of the other matrix are far from this subspace.

Lemma 4.5 (Eigenspace closeness) *Let $X \in \mathbb{R}^{n\times n}$ be a symmetric matrix, and let $v_1, \dots, v_k$ be any k of its eigenvectors with non-zero eigenvalues $\lambda_1, \dots, \lambda_k$. Let $\lambda = \min\{|\lambda_1|, \dots, |\lambda_k|\}$. Also let $Y \in \mathbb{R}^{n\times n}$ be a symmetric matrix, and let $P \in \mathbb{R}^{n\times n}$ be the orthogonal projection to the span of the eigenvectors of Y that are smaller in absolute value than $\lambda - \eta$ for some $\eta > 0$. Then, for any orthonormal* *basis* $u_1, \ldots, u_k \in \mathbb{R}^n$ *of* $\text{span}(v_1, \ldots, v_k)$*,*

$$\sum_{i=1}^{k} Pu_i^2 \leq \frac{1}{\eta^2} \sum_{i=1}^{k} (X - Y)u_i^2 .$$

Proof First, for the eigenvectors $v_1, \ldots, v_k$ of X we have that

$$(X - Y)v_i^2 = \lambda_i v_i - Yv_i^2 = (\lambda_i I_n - Y)v_i^2 \geq P(\lambda_i I_n - Y)v_i^2 . \quad (4.1)$$

Let now $v'_1, \ldots, v'_m \in \mathbb{R}^n$ be the eigenvectors of Y in P with corresponding eigenvalues $\lambda'_1, \ldots, \lambda'_m$ smaller in absolute value than $\lambda - \eta$. Continuing **??**,

$$(X - Y)v_i^2 \geq \sum_{j=1}^{m} (\lambda_i - \lambda'_j)^2 \langle v'_j, v_i \rangle^2 \geq \eta^2 \sum_{j=1}^{m} \langle v'_j, v_i \rangle^2 = \eta^2 Pv_i^2 . \quad (4.2)$$

Second, we claim that $\sum_{i=1}^{k} (X - Y)v_i^2 = \sum_{i=1}^{k} (X - Y)u_i^2$ and $\sum_{i=1}^{k} Pv_i^2 = \sum_{i=1}^{k} Pu_i^2$. The proof is simply that, for any matrix $M \in \mathbb{R}^{n \times n}$ and any two orthonormal bases of the same subspace $\{x_i\}$ and $\{y_i\}$,

$$\sum_i Mx_i^2 = \sum_i (M^\top M x_i x_i^\top) = (M^\top M \sum_i x_i x_i^\top) = (M^\top M \sum_i y_i y_i^\top) ,$$

and by the same sequence of transformations we get back that $(M^\top M \sum_i y_i y_i^\top) = \sum_i My_i^2$. Using this fact for the orthonormal bases $\{v_i\}$ and $\{u_i\}$, we prove the desired fact, and combining it with **??** we complete the proof. □

We state now the well-known bounds of subspace enumeration.

Lemma 4.6 (Subspace enumeration) *There exists an algorithm that outputs with high probability an* δ*-net of the unit* $(d-1)$*-dimensional ball in time* $(1/\delta)^{O(d)}$*.*

Proof By standard arguments, there exists such a cover of size $m = O(1/\delta)^d$ (see Example 5.8 in **?**). Furthermore, it suffices to sample $O(m \log m)$ random points on the unit ball in order to obtain with high probability such a cover.□

Finally, we prove **??**.

Proof (Proof of ??) We have $A\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq \delta$ for all $a \in [k]$, so $\sum_{a=1}^{k} (A - W)\frac{Z_a}{Z_a}^2 \leq \delta k$. Because $\frac{Z_a}{Z_a}$ are orthonormal, for any orthonormal $u_1, \ldots, u_k$ that span the same subspace we also have $\sum_{i=1}^{k} (A - W)u_i^2 \leq \delta k$. Also recall that W has column span in the span of $\{Z_a\}_{a \in [k]}$, so all of its eigenvectors with non-zero eigenvalues lie in the span of $\{u_1, \ldots, u_k\}$. Let P be the orthogonal projection to the eigenvectors of A with eigenvalue smaller than λ in absolute value. Then, by **??** applied with W, A as X, Y, we get for any orthonormal vectors $u_1, \ldots, u_r$ that span the eigenspace of W with eigenvalues larger than $\lambda + \eta$ in absolute value, for any $\eta > 0$, that $\sum_{i=1}^{r} Pu_i^2 \leq \delta k/\eta^2$. Hence, $Pu_i^2 \leq \delta k/\eta^2$

for all $i \in [r]$, so each u_i is $O(\sqrt{\delta k/\eta^2})$-close to the subspace corresponding to $P^\perp = I_n - P$.

Then, by **??**, by finding an $O(\sqrt{\delta k/\eta^2})$-net of the eigenspace of A corresponding to $P^\perp$, we are guaranteed to find a vector that is $O(\sqrt{\delta k/\eta^2})$-close to each u_i. This subspace has dimension at most t, so applying **??** takes time $(\delta k/\eta^2)^{O(t)}$ and produces a list of size at most $(\delta k/\eta^2)^{O(t)}$. The overall time complexity is then bounded by $n^{O(1)} \cdot (\delta k/\eta^2)^{O(t)}$. □

4.1.2 Eigenvector coordinate separation

In this section we prove **??**. First, we state a technical lemma.

Lemma 4.7 (Non-uniformity of eigenvectors) *Let $c \geq 0$, $0 < \lambda < \zeta$, and $\alpha \geq \sqrt{2}c\lambda$ be parameters. Let $D \in \mathbb{R}^k$ with $D_x \leq c$ for all $x \in [k]$. Let M be a symmetric $k \times k$ matrix with $M \leq \zeta$ such that for any two distinct columns M_x and M_y it holds that $D_x M_x - D_y M_y{}^2 \geq \alpha^2$. Let $v_1, \dots, v_k$ be orthonormal eigenvectors of M and let $S := v_r \mid Mv_r = \tau v_r, |\tau| > \lambda$. Then*

$$\min_{x \neq y \in [k]} \max_{v_r \in S} D_x(v_r)_x - D_y(v_r)_y \geq \sqrt{\frac{\alpha^2 - 2c^2\lambda^2}{k(\zeta^2 - \lambda^2)}}\,.$$

Proof We write $M = \sum_{r=1}^k \lambda_r v_r v_r^\top$ and get that, for any $x \neq y \in [k]$,

$$D_x M_x - D_y M_y = \sum_{r=1}^k \lambda_r (D_x(v_r)_x - D_y(v_r)_y) v_r\,.$$

Using that $M \leq \zeta$ and $\lambda_r \leq \lambda$, we get

$$\begin{aligned} D_x M_x - D_y M_y{}^2 &= \sum_{r=1}^k \lambda_r^2 (D_x(v_r)_x - D_y(v_r)_y)^2 \\ &= \sum_{r \in S} \lambda_r^2 (D_x(v_r)_x - D_y(v_r)_y)^2 + \sum_{r \notin S} \lambda_r^2 (D_x(v_r)_x - D_y(v_r)_y)^2 \\ &\leq \zeta^2 \sum_{r \in S} (D_x(v_r)_x - D_y(v_r)_y)^2 + \lambda^2 \sum_{r \notin S} (D_x(v_r)_x - D_y(v_r)_y)^2\,. \end{aligned} \tag{4.3}$$

Because the eigenvectors are orthonormal, the rows of the matrix $V \in \mathbb{R}^{k \times k}$ whose columns are $v_{r r \in [k]}$ are also orthonormal: that is, $\sum_{r=1}^k (v_r)_x (v_r)_y = 0$ for $x \neq y$ and $\sum_{r=1}^k (v_r)_x^2 = 1$ for $x \in [k]$. Therefore, we obtain

$$\begin{aligned} \sum_{r \notin S} (D_x(v_r)_x - D_y(v_r)_y)^2 &= \sum_{r=1}^k (D_x(v_r)_x - D_y(v_r)_y)^2 - \sum_{r \in S} (D_x(v_r)_x - D_y(v_r)_y)^2 \\ &= D_x^2 + D_y^2 - \sum_{r \in S} (D_x(v_r)_x - D_y(v_r)_y)^2 \\ &\leq 2c^2 - \sum_{r \in S} (D_x(v_r)_x - D_y(v_r)_y)^2\,. \end{aligned}$$

Plugging it into **??**, we get

$$D_xM_x - D_yM_y{}^2 \le (\zeta^2 - \lambda^2)\sum_{r\in S}(D_x(v_r)_x - D_y(v_r)_y)^2 + 2c^2\lambda^2\,.$$

Since $D_xM_x - D_yM_y{}^2 \ge \alpha^2$, it follows that

$$\sum_{r\in S}(D_x(v_r)_x - D_y(v_r)_y)^2 \ge \frac{\alpha^2 - 2c^2\lambda^2}{\zeta^2-\lambda^2}\,.$$

Finally, by an averaging argument, there must exist some $r \in S$ such that

$$(D_x(v_r)_x - D_y(v_r)_y)^2 \ge \frac{\alpha^2 - 2c^2\lambda^2}{k(\zeta^2-\lambda^2)}\,.$$

Next, we prove **??**.

Proof (Proof of ??) Observe that if $Mv = \tau v$, then we also have that $B^{1/2}v$ is an eigenvector of $\widetilde{M} = B^{1/2}MB^{-1/2}$ of value τ. Note that $\widetilde{M}$ is symmetric and continues to satisfy $\widetilde{M} \le \zeta$. Then, we apply **??** with the matrix $\widetilde{M}$ and with the eigenvectors $B^{1/2}v$ of $\widetilde{M}$ with eigenvalue at least λ in absolute value (normalized to be unit vectors). We apply the lemma with $D \in \mathbb{R}^k$ such that $D_x = (\sqrt{n}B^{-1/2})_x$, whose largest entry is upper bounded by $1/\sqrt{c}$. Recall that we assume any two rows of M satisfy $M^{(a)} - M^{(b)} \ge \alpha$, so also any two rows of $\sqrt{n}MB^{-1/2} = \sqrt{n}B^{-1/2}\widetilde{M}$ satisfy $\sqrt{n}B^{-1/2}(M^{(a)} - M^{(b)}) \ge \alpha$. Taking the transpose and using that $\widetilde{M}$ is symmetric, any two columns of $\sqrt{n}\widetilde{M}B^{-1/2}$ satisfy $\sqrt{n}B^{-1/2}(a)\widetilde{M}_a - \sqrt{n}B^{-1/2}(b)\widetilde{M}_b \ge \alpha$. Applying **??**, and using that $\alpha^2 \ge 4\lambda^2/c$ and $\zeta^2 \ge 4\lambda^2$, we get

$$\min_{a\neq b\in[k]}\max_{v\in S}\frac{1}{B^{1/2}v}\sqrt{n}v_a - v_b \ge \Omega\Big(\sqrt{\frac{\alpha^2 - 2\lambda^2/c}{k(\zeta^2-\lambda^2)}}\Big) \ge \Omega\Big(\frac{\alpha}{\zeta\sqrt{k}}\Big)\,.$$

The above shows that, for all $a \neq b \in [k]$, there exists $\frac{v}{v} \in S$ with $B^{1/2}v = 1$ such that $\sqrt{n}v_a - v_b \ge \Omega(\frac{\alpha}{\zeta\sqrt{k}})$, so $v_a - v_b \ge \Omega(\frac{\alpha}{\zeta\sqrt{kn}})$. We note that $B^{1/2}v = 1$ implies that $\frac{1}{\sqrt{n}} \le v \le \frac{1}{\sqrt{cn}}$. We also have $\sqrt{cn}\|v\| \le \|Zv\| \le \sqrt{n}\|v\|$, so $\sqrt{c} \le \|Zv\| \le \frac{1}{\sqrt{c}}$. Then, if we consider for each such v some $v' \propto v$ such that $\|Zv'\| = 1$, we have that $v'_a - v'_b \ge \Omega(\frac{\alpha\sqrt{c}}{\zeta\sqrt{kn}})$ $\square$

4.1.3 Clustering with separated rows

In this section we prove **??**. First, we state the guarantees of a clustering algorithm used in the proof.

Lemma 4.8 (Partitioning based on distinguishing eigenvectors (based on Lemma E.7 in ?)) *Let $\chi : [n] \to [k]$ be a k-partition of $[n]$. Suppose $|\chi^{-1}(a)| \geq cn$ for all $a \in [k]$. For $\ell \leq k$, let $v_{ii \in [\ell]}$ be k-dimensional vectors that satisfy*

$$\min_{a \neq b \in [k]} \max_{i \in [\ell]} (v_i)_a - (v_i)_b \geq \frac{\alpha}{\sqrt{n}}.$$

Also let $\hat{u}_{ii \in [\ell]}$ be n-dimensional unit vectors that satisfy $\sum_{j=1}^{\ell} \hat{u}_j - Zv_j^2 \leq \delta$. Then **??** *runs in time $n^{O(1)} \cdot (\alpha\sqrt{c})^{-O(k^2)}$ and returns a list of $(\alpha\sqrt{c})^{-O(k^2)}$ k-partitions $\hat{\chi} : [n] \to [k]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x \sim \mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O(\frac{\delta k}{\alpha^2})$.*

Algorithm 2 Spectral partitioning algorithm (based on Algorithm 10 in **?**)

Guess k-dimensional vectors $\{v_i^*\}_{i \in [\ell]}$ with entries in $[-1/\sqrt{cn}, 1/\sqrt{cn}]$ up to entrywise resolution $\alpha/\sqrt{12kn}$. Let $S_1 = \emptyset, \ldots, S_k = \emptyset$. Insert each $x \in [n]$ in S_a where

$$a =_{a \in [k]} \sum_{j=1}^{\ell} ((\hat{u}_j)_x - (v_j^*)_a)^2.$$

Proof To begin with, we remark that the number of guesses is bounded by

$$((2/\sqrt{cn})/(\alpha/\sqrt{12kn}))^{k\ell} \leq O(\sqrt{k}/(\alpha\sqrt{c}))^{k^2},$$

so the algorithm leads to the desired time complexity bound and list size bound.

Let $u_j = Zv_j$, and note that if $Z_{ia} = 1$ then $(u_j)_i = (v_j)_a$. Define the set

$$\mathrm{Good} := x \in [n] \ \mid \ ((\hat{u}_j)_x - (u_j)_x)^2 \leq \frac{\alpha^2}{12kn}, \forall j \in [\ell].$$

Note that, because $\hat{u}_j - Zv_j^2 \leq \delta$ for all $j \in [\ell]$, we have that $n - |\mathrm{Good}| \leq \frac{\delta}{\alpha^2/(12kn)} = \frac{12\delta k}{\alpha^2} n$, so $|\mathrm{Good}| \geq n - \frac{12\delta k}{\alpha^2} n$.

Consider the guess of the algorithm in which $|(v_i^*)_a - (v_i)_a| \leq \alpha/\sqrt{12kn}$ for all $a \in [k]$, $i \in [\ell]$. Consider some $x \in \mathrm{Good}$ with $Z_{xa} = 1$ for some $a \in [k]$. Then $((\hat{u}_j)_x - (u_j)_x)^2 \leq \frac{\alpha^2}{12kn}$ and $(u_j)_x = (v_j)_a$, so

$$\begin{aligned}
((\hat{u}_j)_x - (v_j^*)_a)^2 &= (((\hat{u}_j)_x - (u_j)_x) + ((u_j)_x - (v_j)_a) + ((v_j)_a - (v_j^*)_a))^2 \\
&= (((\hat{u}_j)_x - (u_j)_x) + ((v_j)_a - (v_j^*)_a))^2 \\
&\leq 2((\hat{u}_j)_x - (u_j)_x)^2 + 2((v_j)_a - (v_j^*)_a)^2 \\
&\leq \frac{\alpha^2}{3kn}.
\end{aligned}$$

Then, overall, we have

$$\sum_{j=1}^{\ell}((\hat{u}_j)_x - (v_j^*)_a)^2 \leq \ell\frac{\alpha^2}{3kn} \leq \frac{\alpha^2}{3n}\,.$$

On the other hand, for any $b \neq a \in [k]$, there exists $i \in [\ell]$ with $(u_i)_x - (v_i)_b = (v_i)_a - (v_i)_b \geq \frac{\alpha}{\sqrt{n}}$, so similarly we have

$$\begin{aligned}
((\hat{u}_i)_x - (v_i^*)_b)^2 &= \Big(((\hat{u}_i)_x - (u_i)_x) + ((u_i)_x - (v_i)_b) + ((v_i)_b - (v_i^*)_b)\Big)^2 \\
&\geq (\frac{\alpha}{\sqrt{n}} - 2\frac{\alpha}{\sqrt{12kn}})^2 \\
&> \frac{\alpha^2}{3n}\,.
\end{aligned}$$

Hence,

$$\sum_{j=1}^{\ell}((\hat{u}_j)_x - (v_j^*)_b)^2 > \frac{\alpha^2}{3n}\,.$$

This shows that there exists a guess of the algorithm $v_1^*, \ldots, v_\ell^*$ for which all vertices in Good get correctly partitioned. Then, because $|\text{Good}| \geq n - \frac{12\delta k}{\alpha^2}n$, for this guess at most $\frac{12k\delta}{\alpha^2}n$ vertices are misclassified. □

Finally, we prove **??**.

Proof (Proof of ??) For each v_i, let $v_i' \propto v_i$ such that $Zv_i' = 1$. Then the assumptions guarantee that

$$\min_{a \neq b \in [k]} \max_{i \in [r]} (v_i')_a - (v_i')_b \geq \frac{\alpha}{\sqrt{n}}\,.$$

We also have that $\hat{u}_i - Zv_i'^2 \leq \delta$ for all $i \in [r]$. Then the proof follows by **??**: we can recover a list of partitions such that one of them contains (up to a permutation) at most $O(\frac{\delta k}{\alpha^2})n$ wrongly partitioned vertices. The time complexity is $n^{O(1)} \cdot (\alpha\sqrt{c})^{-O(k^2)}$ the list size bound is $(\alpha\sqrt{c})^{-O(k^2)}$. □

Chapter 5

Variance Analysis

5.1 Variance of colourings in expander graphs

In this section we prove a structural result for colourable expander graphs concerning the variance of the connections between any two colour classes. This result is essential to our proof of **??**.

Lemma 5.1 (Variance of k-colourings in expanders) *Let G be an n-vertex d-regular graph with adjacency matrix A, and let $\tilde{A} = \frac{1}{d}A$. Let λ_2 be the second largest eigenvalue of $\tilde{A}$. Let $\chi : [n] \to [k]$ be a k-partition of its vertices with* $\min_{a\in[k]} |\chi^{-1}(a)| \geq cn$*. Suppose that G has at most δdn edges with both endpoints in the same part with respect to χ. For $x \in [n]$ and $a \in [k]$, let $xa = \sum_{y\in\chi^{-1}(a)} \tilde{A}_{xy}$. Then, for any $a, b \in [k]$ and x uniformly random in $\chi^{-1}(a)$,*

$$\mathbb{E}[_{]x\sim\chi^{-1}(a)(xb-\mathbb{E}[_{]x\sim\chi^{-1}(a)xb)^2 \leq O(\frac{\lambda_2}{c}+\frac{\delta}{\lambda_2 c})}}.$$

Proof Fix some $a, b \in [k]$. For simplicity, we denote $\sigma^2 = \mathbb{E}[_{]x\sim\chi^{-1}(a)(xb-\mathbb{E}[_{]x\sim\chi^{-1}(a)xb)^2}}$.

If $a = b$, we have

$$\sigma^2 \leq \mathbb{E}[_{]x\sim\chi^{-1}(a)(xa)^2 \leq \mathbb{E}[_{]x\sim\chi^{-1}(a)xa \leq \frac{2\delta n}{|\chi^{-1}(a)|} \leq \frac{2\delta}{c}}}.$$

We focus now on the case $a \neq b$. By regularity, $\lambda_1 = 1$ corresponds to eigenvector . Let $\beta \in [0, 1]$ be some parameter to be chosen later. We define $v \in \mathbb{R}^n$ as follows:

- $v_x := \frac{1}{\sqrt{|\chi^{-1}(a)|}}(xb - \mathbb{E}[_{]x\sim\chi^{-1}(a)xb)}$ for $x \in \chi^{-1}(a)$,
- $v_x := \frac{1}{\sqrt{|\chi^{-1}(a)|}}\beta$ for $x \in \chi^{-1}(b)$,

- $v_x := 0$ otherwise.

Let us calculate $v^\top \tilde{A} v$. Let $\tilde{A}_{(a)} \in \mathbb{R}^{|\chi^{-1}(a)| \times |\chi^{-1}(a)|}$ be the restriction of $\tilde{A}$ to rows and columns in $\chi^{-1}(a)$, and let $v_{(a)} \in \mathbb{R}^{|\chi^{-1}(a)|}$ be the restriction of v to entries in $\chi^{-1}(a)$. Then

$$\begin{aligned} v^\top \tilde{A} v &= v_{(a)}^\top \tilde{A}_{(a)} v_{(a)} + v_{(b)}^\top \tilde{A}_{(b)} v_{(b)} + 2\beta \frac{1}{|\chi^{-1}(a)|} \sum_{x \in \chi^{-1}(a)} \sum_{\substack{y \in \chi^{-1}(b) \\ y \sim x}} (xb - \mathbb{E}[_{]x \sim \chi^{-1}(a) xb}) \\ &= v_{(a)}^\top \tilde{A}_{(a)} v_{(a)} + v_{(b)}^\top \tilde{A}_{(b)} v_{(b)} + 2\beta \mathbb{E}[_{]x \sim \chi^{-1}(a) xb (xb - \mathbb{E}[_{]x \sim \chi^{-1}(a) xb})}\,. \end{aligned} \tag{5.1}$$

For the first two terms, we have

$$v_{(a)}^\top \tilde{A}_{(a)} v_{(a)} \geq -\Big(\sum_{x,y \in \chi^{-1}(a)} \tilde{A}_{xy}\Big) \cdot v_{(a)}{}_\infty^2 \geq -\frac{2\delta n}{|\chi^{-1}(a)|} \geq -\frac{2\delta}{c}$$

and

$$v_{(b)}^\top \tilde{A}_{(b)} v_{(b)} \geq -\Big(\sum_{x,y \in \chi^{-1}(b)} \tilde{A}_{xy}\Big) \cdot v_{(b)}{}_\infty^2 \geq -\frac{2\delta n}{|\chi^{-1}(a)|} \geq -\frac{2\delta}{c}\,.$$

For the cross-term, we use that

$$= 2\beta \mathbb{E}[_{]x \sim \chi^{-1}(a) (xb - \mathbb{E}[_{]x \sim \chi^{-1}(a) xb})^2}$$

= $2\beta\sigma^2$.

$2\beta \mathbb{E}[_{]x \sim \chi^{-1}(a) xb (xb - \mathbb{E}[_{]x \sim \chi^{-1}(a) xb})} \quad = 2\beta \mathbb{E}[_{]x \sim \chi^{-1}(a) (xb - \mathbb{E}[_{]x \sim \chi^{-1}(a) xb})^2 + 2\beta \mathbb{E}[_{]x \sim \chi^{-1}(a) (\mathbb{E}[_{]x \sim \chi^{-1}(a) xb})(xb - \mathbb{E}[_{]x \sim \chi^{-1}(a) xb}) = 2\beta \mathbb{E}[_{]x \sim \chi^{-}}}}$

Then, plugging these bounds back into **??**, we get

$$v^\top \tilde{A} v \geq 2\beta\sigma^2 - 4\delta/c\,.$$

We also note that $\|v\|^2 = \sigma^2 + \frac{|\chi^{-1}(b)|}{|\chi^{-1}(a)|}\beta^2 \leq \sigma^2 + \beta^2/c$ and $\langle v, \frac{}{\sqrt{n}} \rangle^2 = \frac{|\chi^{-1}(b)|^2}{|\chi^{-1}(a)| n}\beta^2 \leq \beta^2/c$, so

$$v^\top \tilde{A} v \leq \frac{1}{n}\langle v, \rangle^2 + \lambda_2\big(\|v\|^2 - \frac{1}{n}\langle v, \rangle^2\big) \leq \lambda_2\sigma^2 + \beta^2/c\,.$$

Putting everything together,

$$2\beta\sigma^2 - 4\delta/c \leq v^\top \tilde{A} v \leq \lambda_2\sigma^2 + \beta^2/c\,,$$

and taking $\beta = \lambda_2$ we get

$$\sigma^2 \leq \lambda_2/c + \frac{4\delta}{\lambda_2 c} \leq O\Big(\frac{\lambda_2}{c} + \frac{\delta}{\lambda_2 c}\Big)\,.$$

Chapter 6

Algorithms

6.1 Finding colourings and independent sets

6.1.1 colourable models

Our most general result for k-colourings is the following:

Theorem 6.1 (Result for k-colouring) *Let $M \in \mathbb{R}_{\geq 0}^{k \times k}$ be a reversible row stochastic matrix with zero on the diagonal and non-positive second eigenvalue, and let π be its stationary distribution. Let $S_1, \ldots, S_{k'}$ be a partition of $[k]$ such that, for all $a, b \in S_i$ the rows $M^{(a)}$ and $M^{(b)}$ of M are equal, and for all $a \in S_i, b \in S_j$ with $i \neq j$ the rows $M^{(a)}$ and $M^{(b)}$ of M are distinct. For all $i \in [k']$, let $a_i^* =_{a \in S_i} \pi(a)$, and let $p_i = \max(0, \pi(a_i^*) - \sum_{a \neq a_i^* \in S_i} \pi(a))$. Then, given a d-regular (M, δ)-colourable λ-one-sided expander G, where $\delta, \lambda > 0$ are small enough, there exists an algorithm that runs in time $n^{O(1)} \cdot O_{\delta,\lambda,M}(1)$ and outputs a $(k', 1 - \sum_{i \in [k']} p_i + O_M(\sqrt{\delta} + \lambda + \delta/\lambda))$-colouring for G.*

As a direct corollary, we approach a full colouring in the case when all rows of M are distinct:

Corollary 6.2 (Result for k-colouring, distinct rows) *Let $M \in \mathbb{R}_{\geq 0}^{k \times k}$ be a reversible row stochastic matrix with zero on the diagonal, non-positive second eigenvalue, and distinct rows. Then, given a d-regular (M, δ)-colourable λ-one-sided expander G, where $\delta, \lambda > 0$ are small enough, there exists an algorithm that runs in time $n^{O(1)} \cdot O_{\delta,\lambda,M}(1)$ and outputs a $(k, O_M(\sqrt{\delta} + \lambda + \delta/\lambda))$-colouring for G.*

We start by proving two lemmas that we will use in the proof. First, we show that the models we consider in **??** are well-behaved: they have a unique stationary distribution in which each vertex has non-zero probability.

Lemma 6.3 (Model colour class size lower bound) *Let $M \in \mathbb{R}_{\geq 0}^{k \times k}$ be a reversible row stochastic matrix with zeros on the diagonal. Then the stationary distribution π of M is unique and satisfies $\min_{a \in [k]} \pi_a \geq \Omega_M(1)$. Furthermore, for every*

$A \in \mathbb{R}^{n\times n}$ and every k-partition $\chi : [n] \to [k]$ of $[n]$ such that $M - M(A,\chi) \le \delta$, then also $\frac{1}{n}|\chi^{-1}(a)| - \pi_a \le 8\sqrt{\delta}k$ for all $a \in [k]$.

Proof First, because M has non-positive second eigenvalue, its non-zero entries describe a connected graph. Furthermore, for a stationary distribution π of M, for all $a \neq b \in [k]$ we have $\pi_a M_{ab} = \pi_b M_{ba}$, so if $M_{ab}, M_{ba} > 0$ also

$$\Omega_M(1) \le \frac{\pi_a}{\pi_b} = \frac{M_{ab}}{M_{ba}} \le O_M(1)\,.$$

Then, because the non-zero entries of M describe a connected graph, we can chain inequalities of this form to obtain for all $a \neq b \in [k]$ the ratios π_a/π_b. Together with the fact that $\sum_{a\in[k]} \pi_a = 1$, we get that π is uniquely determined by M and satisfies $\min_{a\in[k]} \pi_a \ge \Omega_M(1)$.

Second, we have that $M(A,\chi)_{ab}$ is δ-close to M_{ab} for all $a, b \in [k]$. Also, for all $a \neq b \in [k]$ we have $\frac{1}{n}|\chi^{-1}(a)|M(A,\chi)_{ab} = \frac{1}{n}|\chi^{-1}(b)|M(A,\chi)_{ba}$ and $\sum_{a\in[k]} \frac{1}{n}|\chi^{-1}(a)| = 1$. Then, if $M_{ab}, M_{ba} > 0$ and $\delta > 0$ is small enough such that $\delta \le \sqrt{\delta}M_{ab}, \sqrt{\delta}M_{ba}$, also

$$(1-4\sqrt{\delta})\frac{\pi_a}{\pi_b} \le \frac{(1-\sqrt{\delta})M_{ab}}{(1+\sqrt{\delta})M_{ba}} \le \frac{\frac{1}{n}|\chi^{-1}(a)|}{\frac{1}{n}|\chi^{-1}(b)|} = \frac{M(A,\chi)_{ab}}{M(A,\chi)_{ba}} \le \frac{(1+\sqrt{\delta})M_{ab}}{(1-\sqrt{\delta})M_{ba}} \le (1+4\sqrt{\delta})\frac{\pi_a}{\pi_b}\,.$$

This implies that for all $a, b \in [k]$ we have

$$(1-8\sqrt{\delta}k)\frac{\pi_a}{\pi_b} \le (1-4\sqrt{\delta})^{k-1}\frac{\pi_a}{\pi_b} \le \frac{\frac{1}{n}|\chi^{-1}(a)|}{\frac{1}{n}|\chi^{-1}(b)|} \le (1+4\sqrt{\delta})^{k-1}\frac{\pi_a}{\pi_b} \le (1+4\sqrt{\delta}k)\frac{\pi_a}{\pi_b}\,,$$

so summing over all $a \in [k]$ we get

$$(1-8\sqrt{\delta}k)\frac{1}{\pi_b} \le \frac{1}{\frac{1}{n}|\chi^{-1}(b)|} \le (1+4\sqrt{\delta}k)\frac{1}{\pi_b}\,,$$

which finally implies that $\frac{1}{n}|\chi^{-1}(b)| - \pi_b \le 8\sqrt{\delta}k$ for all $b \in [k]$. $\square$

Second, we prove that a k-colourable graph that has small variance as per **??** also has $\tilde{A}$ that is close to $W = ZM(\tilde{A},\chi)B^{-1}Z^\top$ on the indicators of the colour classes.

Lemma 6.4 (Small variance implies close indicators) *Let G be an n-vertex graph with adjacency matrix A, and let $\tilde{A} = \frac{1}{d}A$ for some $d > 0$. Let $\chi : [n] \to [k]$ be a k-partition of its vertices with $\min_{a\in[k]} |\chi^{-1}(a)| \ge cn$, and consider its partition matrices $Z \in \{0,1\}^{n\times k}$ and $B \in \mathbb{R}^{k\times k}$ as per* **??**. *For $x \in [n]$ and $a \in [k]$, let $xa = \sum_{y\in\chi^{-1}(a)} \tilde{A}_{xy}$. Suppose that, for any $a, b \in [k]$ and x uniformly random in $\chi^{-1}(a)$,*

$$\mathbb{E}[_{]x\sim\chi^{-1}(a)(xb-\mathbb{E}[_{]x\sim\chi^{-1}(a)xb)^2\le\delta.Then,for}$$

$W = Z\,M(,\chi)B^{-1}Z^\top$ with $M(\tilde{A},\chi)$ defined as in **??** *with respect to $\tilde{A}$, it holds for all $a \in [k]$ that*

$$\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq \frac{\delta}{c}\,.$$

Proof Note that $W_{xy} = \frac{E(\chi^{-1}(\chi(x)),\chi^{-1}(\chi(y)))}{d\chi^{-1}(\chi(x))\chi^{-1}(\chi(y))}$. Then we get

$$\begin{aligned}
\tilde{A}Z_a - WZ_a^2 &= \sum_{b\in[k]}\sum_{x\in\chi^{-1}(b)}\Big(\sum_{y\in\chi^{-1}(a)}\tilde{A}_{x,y} - \frac{1}{d}\frac{E(b,a)}{|\chi^{-1}(b)|}\Big)^2 \\
&= \sum_{b\in[k]}\sum_{x\in\chi^{-1}(b)}(xa - \mathbb{E}[_{]x\sim\chi^{-1}(b)xa)^2} \\
&= \sum_{b\in[k]}|\chi^{-1}(b)|\;\mathbb{E}[_{]x\sim\chi^{-1}(b)(xa-\mathbb{E}[_{]x\sim\chi^{-1}(b)xa)^2}} \\
&\leq \delta n\,,
\end{aligned}$$

so using that $cn \leq Z_a^2 \leq n$ we get that $\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq \frac{\delta}{c}$. □

Let us now prove **??**.

Proof (Proof of ??) We note by **??** that π is unique and satisfies $\min_{a\in[k]}\pi_a \geq \Omega_M(1)$.

Denote by $\chi : [n] \to [k]$ the k-partition with respect to which G is (M,δ)-colourable. Let $\varphi : [k] \to [k']$ map each $a \in [k]$ to the $i \in [k']$ for which $a \in S_i$, and then define $\chi' : [n] \to [k']$ as $\chi'(x) = \varphi(\chi(x))$ for all $x \in [n]$. That is, χ' is a k'-partition of the graph that starts with the k-partition χ and then for each $i \in [k']$ contracts all colours in S_i into a single colour.

We aim to apply **??** with respect to the k'-partition χ' and approximately recover the partition of the vertices corresponding to χ'. For that, we need to argue that the k'-partition χ' satisfies the conditions of **??**. Afterward, a simple rounding step will lead to the desired guarantee.

Let A be the adjacency matrix of the graph, and let $\tilde{A} = \frac{1}{d}A$. We define the model matrices $M(\tilde{A},\chi)$ and $M(\tilde{A},\chi')$ as in **??**. We will apply **??** with matrices $M(\tilde{A},\chi')$ and $\tilde{A}$ (to play the role of M and A in **??**, respectively).

We will use the following two claims about the relation between the models defined with respect to χ and χ'. First, we relate the entries of $M(\tilde{A},\chi)$ and $M(\tilde{A},\chi')$.

Claim 6.5 (Closeness of $M(\tilde{A},\chi)$ and $M(\tilde{A},\chi')$) *For any $i,j \in [k']$, we have*

$$M(\tilde{A},\chi')_{ij} = \frac{\sum_{b\in S_j}|\chi^{-1}(b)|}{|\chi^{-1}(b_j^*)|}M(\tilde{A},\chi)_{a_i^*b_j^*} \pm O_M(\delta)\,.$$

Proof A calculation shows that, for $i, j \in [k']$,

$$M(\tilde{A}, \chi')_{ij} = \sum_{a \in S_i} \frac{|\chi^{-1}(a)|}{\sum_{a' \in S_i} |\chi^{-1}(a')|} \sum_{b \in S_j} M(\tilde{A}, \chi)_{ab} \,. \tag{6.1}$$

We note that, for any two rows $M^{(a)}$ and $M^{(a')}$ that are identical, we have that $M(\tilde{A}, \chi)^{(a)} - M(\tilde{A}, \chi)^{(a')} \leq O_M(\delta)$. Hence, for any $a, a' \in S_i$, we have that $\sum_{b \in S_j} M(\tilde{A}, \chi)_{ab}$ is $O_M(\delta)$-close to $\sum_{b \in S_j} M(\tilde{A}, \chi)_{a'b}$. Denote $b_i^* = a_i^*$ for all $i \in [k']$. Then, by **??** and the above,

$$\begin{aligned}
M(\tilde{A}, \chi')_{ij} &= \sum_{b \in S_j} M(\tilde{A}, \chi)_{a_i^* b} \pm O_M(\delta) \\
&= \frac{1}{|\chi^{-1}(a_i^*)|} \sum_{b \in S_j} |\chi^{-1}(b)| M(\tilde{A}, \chi)_{b a_i^*} \pm O_M(\delta) \\
&= \frac{\sum_{b \in S_j} |\chi^{-1}(b)|}{|\chi^{-1}(a_i^*)|} M(\tilde{A}, \chi)_{b_j^* a_i^*} \pm O_M(\delta) \\
&= \frac{\sum_{b \in S_j} |\chi^{-1}(b)|}{|\chi^{-1}(b_j^*)|} M(\tilde{A}, \chi)_{a_i^* b_j^*} \pm O_M(\delta) \,,
\end{aligned}$$

where we used that $|\chi^{-1}(a)| M(\tilde{A}, \chi)_{ab} = |\chi^{-1}(b)| M(\tilde{A}, \chi)_{ba}$ for all $a, b \in [k]$. □

Second, let $W = ZM(\tilde{A}, \chi) B^{-1} Z^\top$, with the partition matrices Z, B corresponding to the k-partition χ, and let $W' = Z' M(\tilde{A}, \chi')(B')^{-1}(Z')^\top$, with the partition matrices Z', B' corresponding to the k'-partition χ'. Then we also relate the entries of W and W'.

Claim 6.6 (Closeness of W and W') *For any $x, y \in [n]$, we have*

$$W'_{xy} = W_{xy} \pm O_M(\delta/n) \,.$$

Proof A calculation shows that, for $x, y \in [n]$,

$$W_{xy} = \frac{M(\tilde{A}, \chi)_{\chi(x)\chi(y)}}{|\chi^{-1}(\chi(y))|}$$

and, by **??**,

$$\begin{aligned}
W'_{xy} &= \frac{M(\tilde{A}, \chi')_{\chi'(x)\chi'(y)}}{|(\chi')^{-1}(\chi'(y))|} \\
&= \frac{\frac{\sum_{b \in S_{\chi'(y)}} |\chi^{-1}(b)|}{|\chi^{-1}(b_{\chi'(y)})|} M(\tilde{A}, \chi)_{a_{\chi'(x)} b_{\chi'(y)}}}{\sum_{b \in S_{\chi'(y)}} |\chi^{-1}(b)|} \pm O_M(\delta/n) \\
&= \frac{M(\tilde{A}, \chi)_{a_{\chi'(x)} b_{\chi'(y)}}}{|\chi^{-1}(b_{\chi'(y)})|} \pm O_M(\delta/n) \,.
\end{aligned}$$

In **??** $a^*_{\chi'(x)}$ and $b^*_{\chi'(y)}$ could in fact be any arbitrary colours in $S_{\chi'(x)}$ and $S_{\chi'(y)}$, respectively. Then, letting $a_{\chi'(x)} = \chi(x)$ and $b_{\chi'(y)} = \chi(y)$, we get that $W'_{xy} = W_{xy} \pm O_M(\delta/n)$. □

Let us prove now that we satisfy the conditions of **??** with matrices $M(\tilde{A}, \chi')$ and $\tilde{A}$. First, we have trivially that $(B')^{1/2} M(\tilde{A}, \chi')(B')^{-1/2}$ is symmetric and that $M(\tilde{A}, \chi') = 1$. We also have that $|\chi^{-1}(a)| \geq \Omega_M(n)$ for all $a \in [k]$ by **??**.

Row separation of $M(\tilde{A}, \chi')$. We prove now that the rows of $M(\tilde{A}, \chi')$ are separated. Consider two rows $M(\tilde{A}, \chi')^{(i)}$ and $M(\tilde{A}, \chi')^{(i')}$ with $i \neq i' \in [k']$. Because $M^{(a^*_i)} \neq M^{(a^*_{i'})}$, we have that $M^{(a^*_i)} - M^{(a^*_{i'})} \geq \Omega_M(1)$, so then by the closeness of M and $M(\tilde{A}, \chi)$ we also have that $M(\tilde{A}, \chi)^{(a^*_i)} - M(\tilde{A}, \chi)^{(a^*_{i'})} \geq \Omega_M(1)$. Therefore, there exists some $b \in [k]$ such that $M(\tilde{A}, \chi)_{a^*_i b} - M(\tilde{A}, \chi)_{a^*_{i'} b} \geq \Omega_M(1)$. By **??** this implies that, for $j = \varphi(b)$, we also have $M(\tilde{A}, \chi')_{ij} - M(\tilde{A}, \chi')_{i'j} \geq \Omega_M(1)$. Hence, $M(\tilde{A}, \chi')^{(i)} - M(\tilde{A}, \chi')^{(i')} \geq \Omega_M(1)$.

Closeness of $\tilde{A}$ and W'. By definition, we only consider G and partitions with respect to which it has a vertex cover of all edges with both endpoints in the same part of size at most δn, so the number of edges with both endpoints in the same part is bounded by $\delta d n$. By **??**, the size of each colour class is lower bounded by some $\Omega_M(n)$. Then by **??** and **??** we have that $\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq O_M(\lambda + \frac{\delta}{\lambda})$ for all $a \in [k]$.

By **??**, we have that $W' - W \leq W' - W_F \leq O_M(\delta)$. Then, for each S_i, denoting its elements as $S_i = \{a_1, \ldots, a_\ell\}$, we are interested in the closeness of $\tilde{A}$ and W' with respect to the indicator $\sum_{j=1}^{\ell} Z_{a_j}$. We have

$$(\tilde{A} - W') \sum_{j=1}^{\ell} Z_{a_j} \leq \sum_{j=1}^{\ell} (\tilde{A} - W') Z_{a_j} \leq \sum_{j=1}^{\ell} (\tilde{A} - W) Z_{a_j} + \sum_{j=1}^{\ell} (W' - W) Z_{a_j} .$$

For the first term on the right-hand side, we have that $\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq O_M(\lambda + \frac{\delta}{\lambda})$ for all $a \in [k]$, so using that $Z_a \leq \sqrt{n}$ we can bound

$$\sum_{j=1}^{\ell} (\tilde{A} - W) Z_{a_j} \leq \sqrt{n} \cdot O_M(\sqrt{\lambda + \frac{\delta}{\lambda}}) .$$

For the second term, we have that

$$\sum_{j=1}^{\ell} (W' - W) Z_{a_j} \leq \sum_{j=1}^{\ell} W' - W Z_{a_j} \leq \sqrt{n} \cdot O_M(\delta) .$$

Then, also using that $\sum_{j=1}^{\ell} Z_{a_j} \geq \Omega_M(\sqrt{n})$, we have

$$(\tilde{A} - W')\frac{\sum_{j=1}^{\ell} Z_{a_j}}{\sum_{j=1}^{\ell} Z_{a_j}}^2 \leq O_M(\lambda + \frac{\delta}{\lambda}),$$

so we satisfy the second condition in **??** with constant $O_M(\lambda + \frac{\delta}{\lambda})$.

Low rank of $\tilde{A}$. Because $_{\geq\lambda}(\tilde{A}) \leq 1$, by applying **??** with $\sigma = \lambda$, we get that the bottom threshold rank of $\tilde{A}$ is bounded by $_{\leq -\sqrt{2\lambda}}(\tilde{A}) \leq 1/\lambda^2$. Note that, using that $\lambda \leq 1$, we have that $\lambda \leq \sqrt{2\lambda}$. Then, for the purposes of **??**, we apply the theorem with eigenvalue threshold $\lambda' = \max\{\sqrt{2\lambda}, \Omega_M(1)\}$ for some small enough $\Omega_M(1)$ such that the conditions of the theorem are satisfied, and this allows us to take $t = O(1/(\lambda')^2) = O_M(1)$.

Rounding. Then, by **??** applied with matrices $M(\tilde{A}, \chi')$ and $\tilde{A}$, we can recover in time $n^{O(1)} \cdot O_{\delta,\lambda,M}(1)$ a list of $O_{\delta,\lambda,M}(1)$ k'-partitions $\hat{\chi}' : [n] \to [k']$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim \mathrm{Unif}([n])}[\hat{\chi}'(x) \neq \chi'(x)] \leq O_M(\lambda + \delta/\lambda)$. Let $= O_M(\lambda + \delta/\lambda)$ be equal to this upper bound. At this point, for each candidate k'-partition, we apply the rounding algorithm in **??** on each of its colour classes to obtain k' independent sets, and return the set of k' independent sets that cover most vertices of G.

Let us analyze the approximation guarantee. For the best k'-partition, the rounding algorithm in **??** will remove at most $2n$ vertices due to misclassifications and at most $2\delta n$ vertices due to edges with both endpoints in the same part with respect to the k-partition χ. The rounding algorithm may also need to remove a significant portion of vertices because we recover k'-partitions instead of k-partitions — this is necessary even if the k'-partition χ' is perfectly recovered.

To analyze these removals, let us start by defining $\bar{a}_i^* =_{a\in S_i} |\chi^{-1}(a)|$ for each $i \in [k']$. Then, for each $i \in [k']$, the rounding algorithm in **??** may remove at most
$\min(\sum_{a\in S_i} |\chi^{-1}(a)|, 2\sum_{a\neq a_i^*\in S_i} |\chi^{-1}(a)|)$ vertices from $(\chi')^{-1}(i)$. Then for each $i \in [k']$ the number of surviving vertices is at least $\bar{p}_i = \max(0, |\chi^{-1}(a_i^*)| - \sum_{a\neq a_i^*\in S_i} |\chi^{-1}(a)|)$. Then the algorithm returns a $(k', 1 - \frac{1}{n}\sum_{i\in[k']} \bar{p}_i + 2\delta + 2)$-colouring for G. Finally, by **??**, for $\delta > 0$ small enough, $\frac{1}{n}|\chi^{-1}(a)|$ and π_a are $O(\sqrt{\delta}k)$-close for all $a \in [k]$, so the algorithm also returns a $(k', 1 - \sum_{i\in[k']} p_i + O_M(\sqrt{\delta}) + 2)$-colouring for G, where p_i is defined as in the statement of the theorem. Recalling that $= O_M(\lambda + \delta/\lambda)$ gives the stated result. $\square$

6.1.2 2-colourable models

Assuming the UGC conjecture, **?** showed that it is NP-hard to recover even a small linear-sized independent set from a graph that is almost bipartite. As a $k = 2$ corollary of **??**, we show that assuming expansion additionally, it is possible to recover almost the full bipartition.

Theorem 6.7 (Result for 2-colouring) *Let G be an n-vertex d-regular graph with adjacency matrix A and let $\tilde{A} = \frac{1}{d}A$. Let λ be the second largest eigenvalue of $\tilde{A}$. Let $\chi : [n] \to [2]$ be a 2-colouring of its vertices, and suppose that G has a vertex cover of all monochromatic edges of size at most δn. Then, for $\delta, \lambda > 0$ small enough, there exists an algorithm that runs in time $n^{O(1)} \cdot O_{\delta,\lambda}(1)$ and outputs a $(2, O(\lambda + \delta/\lambda))$-colouring for G.*

Proof For simplicity of notation, let $M = M(\tilde{A}, \chi)$ be defined as in **??** with respect to $\tilde{A}$. We aim again to apply **??** with matrices M and $\tilde{A}$, and we verify that its conditions hold.

By regularity, the graph can have no independent set containing more than half of the vertices. Hence, $\chi^{-1}(a) \geq (1/2 - \delta)n$ for all $a \in [2]$. Therefore, the diagonal entries of M are at most $\frac{\delta}{1/2-\delta} = O(\delta)$, so by the row stochasticity of M, we have that the two rows have separation $1 - O(\delta)$.

For the rest of the proof, we will largely repeat the analysis in the proof of **??**. Let $W = ZMB^{-1}Z^{\top}$, with the partition matrices Z, B corresponding to the bipartition χ. We have trivially that $B^{1/2}MB^{-1/2}$ is symmetric and that $M = 1$. For the closeness of $\tilde{A}$ and W, we have by **??** and **??** that $\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq O(\lambda + \delta/\lambda)$ for all $a \in [2]$. For the low rank of $\tilde{A}$, we have by applying **??** with $\sigma = \lambda$ that $_{\leq -\sqrt{2\lambda}}(\tilde{A}) \leq 1/\lambda^2$, and also $\lambda \leq \sqrt{2\lambda}$.

Then we apply **??** with matrices M and $\tilde{A}$ and eigenvalue threshold $\lambda' = \sqrt{2\lambda}$, such that it satisfies the condition of the theorem, and such that we can take $t = O(1/(\lambda')^2) = O(1)$. Then, we can recover in time $n^{O(1)} \cdot O_{\delta,\lambda}(1)$ a list of $O_{\delta,\lambda}(1)$ bipartitions $\hat{\chi} : [n] \to [2]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x \sim \text{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O(\lambda + \delta/\lambda)$. As in the proof of **??**, for each candidate bipartition we apply the rounding algorithm in **??** on each of its colour classes to obtain 2 independent sets, and return the set of 2 independent sets that cover most vertices of G. The rest of the analysis is identical to that in the proof of **??**. $\square$

6.1.3 3-colourable models

For the case $k = 3$, it turns out that the model has non-positive eigenvalue and distinct rows whenever its stationary distribution is strictly bounded pointwise by $1/2$.

The first observation is that we can compute explicitly the entries of M in terms of the stationary distribution.

Lemma 6.8 (Model entries for 3-colourings) *Let $M \in \mathbb{R}^{3\times 3}_{\geq 0}$ be a reversible row stochastic matrix with zeros on the diagonal and stationary distribution π. Then for all $a \neq b \in [3]$*

$$M_{ab} = \frac{(2\pi(a) + 2\pi(b)) - 1}{2\pi(a)}.$$

Proof For all $a \in [3]$, we have $\sum_{b\in[3]} M_{ab} = 1$. Also for all $b \in [3] \setminus a$, we have $\pi(a)M_{ab} = \pi(b)M_{ba}$. Hence, taking the off-diagonal entries of the M matrix as unknown, we have 6 variables and 6 independent linear equations, so there is a unique solution. Using $\sum_{a\in[3]} \pi(a) = 1$, for $a \neq b \neq c \in [3]$ we can express it as

$$M_{ab} = \frac{(2\pi(a) + 2\pi(b)) - 1}{2\pi(a)}.$$

We then use this observation to deduce that, whenever the largest probability of a colour class is upper bounded by $1/2 - \gamma$, any two distinct rows of M have separation γ.

Lemma 6.9 (Row separation for 3-colourings) *Let $M \in \mathbb{R}^{3\times 3}_{\geq 0}$ be a reversible row stochastic matrix with zero on the diagonal and stationary distribution π that satisfies $\max_{a\in[3]} \pi_a \leq 1/2 - \gamma$. Then any two distinct rows $M^{(a)}$ and $M^{(b)}$ of M satisfy $M^{(a)} - M^{(b)} \geq \gamma$.*

Proof It suffices to prove that all non-diagonal entries of M are bounded away from 0. From **??**, we get for $a \neq b$ that $M_{ab} = \frac{(2\pi_a + 2\pi_b) - 1}{2\pi_a}$. Because $\max_{a\in[3]} \pi_a \leq 1/2 - \gamma$, it follows that $\pi_a + \pi_b \geq 1/2 + \gamma$, so $M_{ab} \geq \frac{\gamma}{\pi_a} \geq \gamma$. Then the conclusion follows because $M^{(a)} - M^{(b)} \geq M_{ab} - M_{bb} = M_{ab} \geq \gamma$. $\square$

Finally, we state and prove our result specialized to 3-colourings.

Theorem 6.10 (Result for 3-colouring) *Let G be an n-vertex d-regular graph with adjacency matrix A, and let $\tilde{A} = \frac{1}{d}A$. Let λ be the second largest eigenvalue of $\tilde{A}$. Let $\chi : [n] \to [3]$ be a 3-partition of its vertices, and suppose that G has a vertex cover of all edges with both endpoints in the same part of size at most δn with respect to χ. Suppose $\max_{a\in[3]} \frac{1}{n}|\chi^{-1}(a)| \leq 1/2 - \gamma$. Then, for $\delta, \gamma, \lambda > 0$ small enough, there exists an algorithm that runs in time $n^{O(1)} \cdot O_{\delta,\gamma,\lambda}(1)$ and outputs a $(3, O(\lambda/\gamma^8 + \delta/(\lambda\gamma^8)))$-colouring for G.*

Proof For simplicity of notation, let $M = M(\tilde{A}, \chi)$ be defined as in **??** with respect to $\tilde{A}$. We aim again to apply **??** with matrices M and $\tilde{A}$, and we verify that its conditions hold.

Claim 6.11 *If $\delta \ll \gamma^2$, then any two rows $M^{(a)}$ and $M^{(b)}$ of M satisfy $M^{(a)} - M^{(b)} \geq \Omega(\gamma)$.*

Proof We first argue that M is close to a reversible row stochastic matrix with zero on the diagonal, to which we will apply **??**. Let $M' \in \mathbb{R}^{3\times 3}$ be obtained from M by setting the diagonal entries to zero and by rescaling each row such that it sums up to 1. Then M' is row stochastic with zero on the diagonal. Let us now understand the stationary distribution of M'. Let π be the stationary distribution of M, which coincides with $\frac{1}{n}|\chi^{-1}(a)|_{a\in[3]}$. Then $\pi_a M_{ab} = \pi_b M_{ba}$ for all $a \neq b \in [3]$, so by construction also $\pi_a(\sum_{c\neq a} M_{ac})M'_{ab} = \pi_b(\sum_{c\neq b} M_{bc})M'_{ba}$ for all $a \neq b \in [3]$. Therefore, the stationary distribution of M' has probability mass function proportional to $\pi_a(\sum_{c\neq a} M_{ac})_{a\in[3]}$, normalized to sum up to 1. Finally, we note that $\max_{a\in[3]} M_{aa} \leq O(\delta/\gamma)$, which also implies that $\sum_{c\neq a} M_{ac} \geq 1 - O(\delta/\gamma)$. Then $M - M' \leq O(\delta/\gamma)$, and also the stationary distribution of M' has maximum probability mass bounded by $1/2 - \gamma + O(\delta/\gamma) \leq 1/2 - \Omega(\gamma)$. By **??**, any two distinct rows of M' have separation $\Omega(\gamma)$. Then, by closeness of M and M', we also have that any two rows of M have separation $\Omega(\gamma) - O(\delta/\gamma) \geq \Omega(\gamma)$. $\square$

Note that we can assume that $\delta \ll \gamma^2$, as the guarantees of the theorem are vacuous otherwise, so we can apply **??**. For the rest of the proof, we will largely repeat the analysis in the proof of **??**, but using that any two distinct rows of M are separated and taking into account explicitly the parameter γ. Let $W = ZMB^{-1}Z^{\top}$, with the partition matrices Z, B corresponding to the 3-partition χ. We have trivially that $B^{1/2}MB^{-1/2}$ is symmetric and that $M = 1$. We also have that $|\chi^{-1}(a)| \geq 2\gamma n$ for all $a \in [3]$.

By **??** we have that any two distinct rows of M have separation $\Omega(\gamma)$. For the closeness of $\tilde{A}$ and W, we have by **??** and **??** that $\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 \leq O(\lambda/\gamma^2 + \frac{\delta}{\gamma^2\lambda})$ for all $a \in [3]$. For the low rank of $\tilde{A}$, we have by applying **??** with $\sigma = \lambda$ that ${}_{\leq -\sqrt{2\lambda}}(\tilde{A}) \leq 1/\lambda^2$, and also $\lambda \leq \sqrt{2\lambda}$.

Then we apply **??** with matrices M and $\tilde{A}$ and eigenvalue threshold $\lambda' = \max\{\sqrt{2\lambda}, \Omega(\gamma^{3/2})\}$, such that it satisfies the condition of the theorem, and such that we can take $t = O(1/(\lambda')^2) = O(1/\gamma^3) = O_\gamma(1)$. Then, we can recover in time $n^{O(1)} \cdot O_{\delta,\gamma,\lambda}(1)$ a list of $O_{\delta,\gamma,\lambda}(1)$ 3-partitions $\hat{\chi} : [n] \to [3]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim\mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O(\lambda/\gamma^8 + \delta/(\lambda\gamma^8))$. As in the proof of **??**, for each candidate 3-partition we apply the rounding algorithm in **??** on each of its colour classes to obtain 3 independent sets, and return the set of 3 independent sets that cover most vertices of G. The rest of the analysis is identical to that in the proof of **??**. $\square$

We also provide a completely different view on the 3-colouring case, highlighting a potentially deeper connection. Theorem 1.1 of **?** shows a general result: given a satisfiable CSP with constraints over the edges of a graph with low *top* threshold rank, a semidefinite programming hierarchy rounding algorithm can efficiently find an assignment to the vertices safisfying most of the constraints. As the colouring constraint is a special case of this, the result implies that a k-colouring violating only a small fraction of edges can be recovered efficiently for any k – what is more, the algorithm doesn't use the bottom eigenspace.

However, we are after small vertex violation, not small edge violation: we want that the number of vertices that are not coloured correctly is small. For $k = 3$ we show a surprising connection: under one-sided expansion and balancedness, low edge-violation implies low vertex violation – hence, an alternative of **??** immediately follows from Theorem 1.1 of **?**. For convenience, we show the contrapositive instead: high vertex violation (i.e. low agreement) implies high edge violation.

Theorem 6.12 (Vertex vs edge violation) *Let $0 < \gamma, \delta < \frac{1}{6}$ be constants. Let G be an n-vertex d-regular graph with $\lambda_2 \leq \frac{\delta^6\gamma^9}{10^2}$ that admits a χ such that $\max_{a\in[3]} \frac{1}{n}|\chi^{-1}(a)| \leq 1/2 - \gamma$. Let χ' be an alternative (not necessarily proper) with agree$(\chi, \chi') \leq 1 - \delta$. Then, χ' must have at least $\frac{1}{10^4}\delta^3\gamma^4 nd$ monochromatic edges.*

The proof is quite calculation heavy and the calculations are similar to those in **??** so we only provide a sketch highlighting the main ideas.

Proof (Proof sketch of ??) Let $\psi^{-1}(a, b) := \chi^{-1}(a) \cap \chi'^{-1}(b)$ be the intersection of the colour classes of χ and χ' – without loss of generality, we can assume that the identity permutation maximises agreement.

The proof starts with a technical lemma showing that as χ and χ' have low agreement, there exist colours a and b such that $\chi'^{-1}(b)$, $\psi^{-1}(a, b)$, and $\psi^{-1}(b, b)$ are not too large and not too small either.

By **??**, we have that variance of the degrees between $\chi^{-1}(a)$ and $\chi^{-1}(b)$ is small. On the other hand, note that if the average degree between $\psi^{-1}(a, b)$ and $\chi'^{-1}(b)$ was much less than the average degree between $\chi^{-1}(a)$ and $\chi^{-1}(b)$, it would imply that the variance of the degrees between $\chi^{-1}(a)$ and $\chi^{-1}(b)$ is large.

However, if χ' had only few monochromatic edges, then the average degree between $\psi^{-1}(a, b)$ and $\psi^{-1}(b, b)$ must be small, so the only way the average degree between $\psi^{-1}(a, b)$ and $\chi'^{-1}(b)$ can be large enough is if much more edges from $\psi^{-1}(a, b)$ go to $\chi'^{-1}(b) \setminus \psi^{-1}(b, b)$ than to $\psi^{-1}(b, b)$.

In this case, however, for suitably chosen α and β, we can construct a test vector v (orthogonal to) showcasing that expansion must be poor by lower bounding $\frac{v^\top A_G v}{v^\top v}$ and using the Courant-Fischer Theorem:

- for $x \in \psi^{-1}(a, b)$, let $v_x := 1$

- for $x \in \psi^{-1}(b,b)$, let $v_x := -\alpha$
- for $x \in \chi'^{-1}(b) \setminus \psi^{-1}(b,b)$, let $v_x := \beta$
- otherwise, let $v_x := 0$

This is a contradiction, so χ' must have many monochromatic edges. $\square$

6.1.4 Random planting

Instead of a regular k-colourable one-sided expander graph, in this section we consider randomly planting a in a regular one-sided expander graph. To aid presentation, let us first define the model and some of the notation that we will use throughout this section.

Definition 6.13 (Randomly planted) *Let H be an n-vertex d-regular graph. Then we say the graph G is obtained by randomly planting a in H if it is sampled according to the following procedure:*

1. *Sample $\chi : [n] \to [k]$ by sampling $\chi(x) \sim \mathrm{Unif}([k])$ independently for each $x \in [n]$,*
2. *Let G be a copy of H, and then remove from G each edge $\{u, v\}$ for which $\chi(u) = \chi(v)$.*

The following is our main recovery result. We note that in **??**, for the case when the host graph is a one-sided expander, we give an algorithm that recovers a colouring on *all* vertices of the graph.

Theorem 6.14 (Result for random planting, partial recovery) *Let H be an n-vertex d-regular graph with adjacency matrix A_H. Suppose ${}_{\geq\lambda}(\frac{1}{d}A_H) \leq t$ for some $0 < \lambda < 1$ and $t \in \mathbb{N}$. Let G be the graph obtained by randomly planting a in H, and let $\chi : [n] \to [k]$ be the associated . Then, for $\lambda > 0$ small enough, there exists an algorithm that runs in time $n^{O(1)} \cdot (O_k(1))^t$ and outputs with high probability a $(k, O_k(1/d^{\Omega(1)}))$-colouring for G.*

To prove **??**, we first prove some properties of the graph G obtained after randomly planting a in H.

Lemma 6.15 (colour class concentration in random planting) *Let H be an n-vertex graph. Let G be the graph obtained by randomly planting a in H for some $k = O(1)$, and let $\chi : [n] \to [k]$ be the associated . Then, with high probability, $|\chi^{-1}(a)| = \frac{n}{k} \pm O(\sqrt{n \log n})$ for all $a \in [k]$.*

Proof The proof follows straightforwardly from a Hoeffding bound and a union bound. $\square$

Lemma 6.16 (Model entry concentration in random planting) *Let H be an n-vertex d-regular graph. Let G be the graph obtained by randomly planting a in*

H for some $k = O(1)$, and let $\chi : [n] \to [k]$ be the associated . Let $A \in \mathbb{R}^{n \times n}$ be the adjacency matrix of G, and let $\tilde{A} = \frac{k}{(k-1)d} A$. Let $M(\tilde{A}, \chi)$ be defined as in **??**. *Then, with high probability, $M_{ab} = \frac{1}{k-1} \pm o(1)$ for all $a \neq b \in [k]$.*

Proof Let $d' = \frac{(k-1)d}{k}$, and for $x \in [n]$ and $a \in [k]$, let $xa = \sum_{y \in \chi^{-1}(a)} \tilde{A}_{xy}$. By linearity of expectation,

$$\mathbb{E}[] \sum_{x \in \chi^{-1}(a)} xb = \frac{1}{d'} \mathbb{E}[] \sum_{x \sim y \in E(H)} ([\chi(x) = a \wedge \chi(y) = b] + [\chi(x) = b \wedge \chi(y) = a])$$

$$= \frac{1}{d'} \frac{nd}{2} \frac{2}{k^2} = \frac{n}{k(k-1)} .$$

By the d-regularity of H, changing the colour of a single vertex can increase or decrease $\sum_{x \in \chi^{-1}(a)} xb$ by at most $O(1)$. Hence, McDiarmid's inequality gives that $\sum_{x \in \chi^{-1}(a)} xb$ is with probability $O(n^{-2})$ within $O(\sqrt{n \log n})$ of its expectation. Using **??** and putting it all together, we get

$$M_{ab} = \frac{\frac{n}{k(k-1)} \pm O(\sqrt{n \log n})}{\frac{n}{k} \pm O(\sqrt{n \log n})} = \frac{1}{k-1} \pm O(\sqrt{\frac{\log n}{n}})$$

with high probability for all $a \neq b \in [k]$. $\square$

Next, we show that with high probability the graph G has a good variance bound.

Lemma 6.17 (Variance of random planting) *Let H be an n-vertex graph. Let G be the graph obtained by randomly planting a in H for some $k = O(1)$, and let $\chi : [n] \to [k]$ be the associated . Let $A \in \mathbb{R}^{n \times n}$ be the adjacency matrix of G, and let $\tilde{A} = \frac{k}{(k-1)d} A$. For $x \in [n]$ and $a \in [k]$, let $xa = \sum_{y \in \chi^{-1}(a)} \tilde{A}_{xy}$. Then, with high probability, for any $a, b \in [k]$ and x uniformly random in $\chi^{-1}(a)$,*

$$\mathbb{E}[_{]x \sim \chi^{-1}(a)(xb - \mathbb{E}[_{]x \sim \chi^{-1}(a)xb)^2 \leq 1/d^{\Omega(1)}}} .$$

Proof If $a = b$, we have $\mathbb{E}[_{]x \sim \chi^{-1}(a)(xb - \mathbb{E}[_{]x \sim \chi^{-1}(a)xb)^2 = 0}}$ by construction, so we assume $a \neq b$.

Writing $xb - \mathbb{E}[_{]x \sim \chi^{-1}(a)xb}$ as $(xb - \frac{1}{k-1}) + (\frac{1}{k-1} - \mathbb{E}[_{]x \sim \chi^{-1}(a)xb)}$ and using that $(a+b)^2 \leq 2(a^2 + b^2)$, we get

$$\mathbb{E}[_{]x \sim \chi^{-1}(a)} \left(xb - \mathbb{E}[_{]x \sim \chi^{-1}(a)xb} \right)^2 \leq 2 \left(\mathbb{E}[_{]x \sim \chi^{-1}(a)} \left(xb - \frac{1}{k-1} \right)^2 + \left(\frac{1}{k-1} - M_{ab} \right)^2 \right) . \quad (6.2)$$

Let us define $S = \sum_{x \in \chi^{-1}(a)} (xb - \frac{1}{k-1})^2$ to analyze the first term. Let $d' = \frac{(k-1)d}{k}$. For any $x \in V(G)$, conditioned on $x \in \chi^{-1}(a)$, by d-regularity we have $xb \sim \frac{1}{d'}\mathrm{Bin}(d, \frac{1}{k})$. Hence, by the law of total probability and linearity of expectation, we get

$$\mathbb{E}[_{]\chi S = \frac{n}{k} \frac{1}{d'^2} d \frac{1}{k}(1-\frac{1}{k}) = \frac{n}{(k-1)kd} \leq O(n/d)}.$$

It is easy to verify that changing the colour of a single vertex can increase or decrease S by at most $O(1)$. Therefore, by McDiarmid's inequality,

$$\Pr \left|S - \mathbb{E}l[]\right| \geq \sqrt{n \log n} \;\leq\; 2\exp(-\Omega(\frac{n \log n}{n})) = o(1).$$

Hence, with high probability

$$S = \mathbb{E}l[\pm] O(\sqrt{n \log n}) = O(n/d) \pm O(\sqrt{n \log n}).$$

Then, using **??**, with high probability

$$\mathbb{E}[_{]x \sim \chi^{-1}(a)(xb - \frac{1}{k-1})^2 = \frac{S}{|\chi^{-1}(a)|} = O(1/d) + O(\sqrt{\frac{\log n}{n}}) = 1/d^{\Omega(1)}}.$$

By **??**, the second term of **??** is also bounded by $(\frac{1}{k-1} - M_{ab})^2 \leq 1/d^{\Omega(1)}$, finishing the proof. $\square$

The last ingredient to prove **??** is a lemma that shows how low threshold rank is approximately preserved after planting.

Lemma 6.18 (Threshold rank kept after planting) *Let H be an n-vertex graph with adjacency matrix A_H. Let G be the graph obtained by planting a in H (not necessarily randomly), and let A_G be its adjacency matrix. Let $\bar{A}_H := \frac{1}{c}A_H$ and $\bar{A}_G := \frac{1}{c}A_G$ be adjacency matrices of H and G respectively normalized by the same parameter $c > 0$. For $r_1, r_2 \geq 0$ and $t_1, t_2 \geq 1$, assuming that*

$$_{\geq r_1}(\bar{A}_H) \leq t_1 \quad \textit{and} \quad _{\leq -r_2}(\bar{A}_H) \leq t_2,$$

it holds that

$$_{\leq -(r_1+r_2)}(\bar{A}_G) \;\leq\; k\,t_1 + t_2, \quad _{\geq r_1+r_2}(\bar{A}_G) \;\leq\; t_1 + k\,t_2.$$

Proof Decompose $\bar{A}_G = \bar{A}_H - \bar{A}_F$, where $\bar{A}_F$ is block-diagonal over the k colour classes. By Cauchy's interlacing theorem, each block has $\leq t_1$ eigenvalues $\geq r_1$ and $\leq t_2$ eigenvalues $\leq -r_2$, so $_{\leq -r_1}(-\bar{A}_F) =_{\geq r_1} (\bar{A}_F) \leq k\,t_1$ and $_{\geq r_2}(-\bar{A}_F) =_{\leq -r_2} (\bar{A}_F) \leq k\,t_2$. Hence, applying Weyl's inequalities

$$_{\leq -(r_1+r_2)}(\bar{A}_H - \bar{A}_F) \leq_{\leq -r_2} (\bar{A}_H) +_{\leq -r_1} (-\bar{A}_F)\,,$$
$$_{\geq r_1+r_2}(\bar{A}_H - \bar{A}_F) \leq_{\geq r_1} (\bar{A}_H) +_{\geq r_2} (-\bar{A}_F)\,,$$

we get the two claimed bounds. $\square$

We are now ready to prove **??**.

Proof (Proof of ??) Let $A \in \mathbb{R}^{n\times n}$ be the adjacency matrix of G, and let $\tilde{A} = \frac{k}{(k-1)d}A$. Also let $d' = \frac{(k-1)d}{k}$. Let $M = M(\tilde{A}, \chi)$ be defined as in **??**. We aim again to apply **??** with matrices M and $\tilde{A}$, and we verify that its conditions hold.

Let $W = ZMB^{-1}Z^{\top}$, with the partition matrices Z, B corresponding to the k-partition χ. We have trivially that $B^{1/2}MB^{-1/2}$ is symmetric. By **??** and the triangle inequality, $M \leq \frac{1}{k-1}(-) + M - \frac{1}{k-1}(-) = 1 + o(1)$, where $\in \mathbb{R}^{k\times k}$ is the matrix with all entries equal to 1. We also have by **??** that with high probability $|\chi^{-1}(a)| \geq (1/k - o(1))n$ for all $a \in [k]$.

In addition, as the diagonal entries of M are zero and by **??** the off-diagonal entries are $\frac{1}{k-1} \pm o(1)$, we have $M^{(a)} - M^{(b)} \geq 2/k$ for any two rows $M^{(a)}$ and $M^{(b)}$ of M. For the closeness of $\tilde{A}$ and W, because of **??** and **??**, we can apply **??** and get that, for all $a \in [k]$,

$$\tilde{A}\frac{Z_a}{Z_a} - W\frac{Z_a}{Z_a}^2 = \frac{1}{d^{\Omega(1)}(\frac{1}{k} - o(1))} \leq O_k(1/d^{\Omega(1)}) \,.$$

For the low rank of $\tilde{A}$, we have ${}_{\geq\lambda}(\frac{1}{d}A_H) \leq t$ by assumption, and as $\frac{1}{d}A_H \leq 1$ because of d-regularity, we can apply **??** with $\sigma = \lambda$ to get ${}_{\leq -\sqrt{2\lambda}}(\frac{1}{d}A_H) \leq t/\lambda^2$ as well. Note that, using that $\lambda \leq 1$, we have that $\lambda \leq \sqrt{2\lambda}$. Let $\lambda' = \max\{\sqrt{2\lambda}, \Omega_k(1)\}$ for some small enough $\Omega_k(1)$, such that ${}_{\geq\lambda'}(\frac{1}{d}A_H) = O_k(t)$ and ${}_{\leq-\lambda'}(\frac{1}{d}A_H) = O_k(t)$. By applying **??** with normalization parameter d, we get ${}_{\geq 2\lambda'}(\frac{1}{d}A_G) = O_k(t)$ and ${}_{\leq -2\lambda'}(\frac{1}{d}A_G) = O_k(t)$. As $k \geq 3$, we have $\frac{d}{d'}2\lambda' \leq 3\lambda'$, so ${}_{\geq 3\lambda'}(\tilde{A}) +_{\leq -3\lambda'} (\tilde{A}) = O_k(t)$.

Then, by the above, with high probability we satisfy the conditions necessary to apply **??** with $1/k - o(1)$, $1 + o(1)$, $2/k$, $O_k(1/d^{\Omega(1)})$, $O(t)$ and $O(\lambda')$ in the roles of c, ζ, α, δ, t and λ, respectively. Then, we can recover in time $n^{O(1)} \cdot (O_k(1))^t$ a list of $(O_k(1))^t$ k-partitions $\hat{\chi} : [n] \to [k]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim \mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O_k(1/d^{\Omega(1)})$. Finally, as in the proof of **??**, for each candidate k-partition we apply the rounding algorithm in **??** on each of its colour classes to obtain k independent sets, and return the set of k independent sets that cover most vertices of G. The rest of the analysis is identical to that in the proof of **??**. $\square$

Rounding to a full colouring

Next, we show that if H actually is a one-sided expander (instead of having just low top threshold rank), and if the degree d is at least a large enough constant, then we can actually obtain a full by exploiting the full randomness of the model.

We start from the guarantees of **??**, which is a version of **??** that omits the last rounding step: instead, it guarantees the recovery of a *list* of candidate k-colourings, such that at least one of them is close to the planted k-colouring. We remark that **??** itself does not guarantee that the output k-colouring is close to the planted k-colouring, while our strategy for obtaining a full colouring relies on this property — hence our use of **??**. Then, we show how to convert this list of partial colourings to a full colouring. Our main result is the following:

Theorem 6.19 (Result for random planting, full recovery) *Let H be an n-vertex d-regular graph with adjacency matrix A_H. Let λ be the second largest eigenvalue of $\frac{1}{d}A_H$. Let G be the graph obtained by randomly planting a in H, and let $\chi : [n] \to [k]$ be the associated . Then, for $\lambda > 0$ small enough and d at least a large enough constant compared to k, there exists an algorithm that runs in time $n^{O(1)} \cdot O_k(1)$ and outputs with high probability a $(k, 0)$-colouring for G.*

The proof and algorithm for **??** adapt Theorem B.10 of **?**. For full details, see **??**; below we outline the key modifications.

The key difference between our setting and that of **?** is that they assume two-sided expansion, while we only assume one-sided expansion. The two ways in which Theorem B.10 of **?** uses two-sided expansion are via the Expander Mixing Lemma (EML) (used in Lemma B.17, Lemma B.19, and Lemma B.20 of **?**)

$$e(S,T) \leq d|S|(\frac{|T|}{n} + c\sqrt{\frac{|T|}{|S|}})$$

and via the vertex expansion lower bound (used in Lemma B.21 of **?**): for $|S| = \alpha n$,

$$N(S) \setminus S \geq (\frac{1}{(1-\alpha)c^2 + \alpha} - 1)|S|\,.$$

For the former, we note that Lemmas B.19 and B.20 of **?** use the EML with $S = T$, and that we can avoid Lemma B.17 of **?** by assuming a large enough (but still constant) d. As the lemma below shows, for $S = T$, the EML holds even under one-sided expansion:

Lemma 6.20 (Edge density upper bound) *Let G be an n-vertex d-regular graph with adjacency matrix A satisfying $\lambda_2(\frac{1}{d}A) \leq c$. Then, for any set $S \subseteq [n]$,*

$$E(G_S) \leq \frac{d|S|}{2}(\frac{|S|}{n} + c)\,,$$

where $E(G_S)$ is the number of edges in the graph induced by S.

For the latter use of two-sided expansion, we can show under one-sided expansion a bound that is only slightly weaker, and suffices for our purposes by assuming slightly stronger expansion:

Lemma 6.21 (Vertex expansion lower bound) *Let G be an n-vertex d-regular graph with adjacency matrix A satisfying $\lambda_2(\frac{1}{d}A) \leq c$. Then, for any set $S \subseteq [n]$ with $|S| \leq \alpha n$,*

$$N(S) \setminus S \geq \left(\frac{1}{c + \sqrt{\alpha}} - 1\right)|S|,$$

where $N(S)$ is the number of vertices outside S adjacent to at least one vertex in S.

The proofs of **??** and **??** are also in **??**.

6.1.5 Independent sets

We state now our result for independent sets.

Theorem 6.22 (Independent sets, small bottom rank) *Let G be an n-vertex d-regular graph with adjacency matrix A, and let $\tilde{A} = \frac{1}{d}A$. For $0 < \gamma < 1/4$, suppose G contains an independent set $I \subseteq [n]$ of size $(\frac{1}{2} - \gamma)n$. Suppose ${}_{\leq -\lambda}(\tilde{A}) \leq t$ for some $0 < \lambda < 1$ and $t \in \mathbb{N}$. Then there exists an algorithm with runtime $(n) \cdot (\gamma/(1-\lambda))^{-O(t)}$ that returns an independent set of size at least $(1/2 - \gamma - (4+c)\gamma/(1-\lambda))n$ for any constant $c > 0$.*

By **??**, we immediately get a result under the assumption that the top threshold rank is small.

Corollary 6.23 (Independent sets, small top rank) *Let G be an n-vertex d-regular graph with adjacency matrix A, and let $\tilde{A} = \frac{1}{d}A$. For $0 < \gamma < 1/4$, suppose G contains an independent set $I \subseteq [n]$ of size $(\frac{1}{2} -)n$. Suppose ${}_{\geq \lambda}(\tilde{A}) \leq t$ for some $0 < \lambda < 1$ and $t \in \mathbb{N}$. Then there exists an algorithm with runtime $(n) \cdot (\gamma/(1-\lambda))^{-O(t)}$ that returns an independent set of size at least $(1/2 - \gamma - (4+c)\gamma/(1-\sqrt{\lambda}))n$ for any constant $c > 0$.*

Proof Because ${}_{\geq \lambda}(\tilde{A}) \leq t$, we get by applying **??** with $\sigma = c$ that ${}_{\leq -\lambda'}(\tilde{A}) \leq t/c^2$ for any constant $c > 0$, where $\lambda' = \sqrt{\lambda(1-c)+c} \geq \sqrt{\lambda}$. The result then follows by **??**. $\square$

To prove **??**, we first prove that the indicator vector of the independent set is close to the subspace spanned by the bottom t eigenvectors of the adjacency matrix:

Lemma 6.24 (Closeness to bottom eigenvectors) *In the same setting as **??**, let $u \in \mathbb{R}^n$ have $u_x = \frac{1}{\sqrt{n}}$ for $x \in I$ and $u_x = -\frac{1}{\sqrt{n}}$ for $x \notin I$. Then, there exists a vector $\tilde{u} \in \mathbb{R}^n$ with $\|\tilde{u}\| \leq \|u\|$ and $||\tilde{u} - u||^2 \leq \frac{4\gamma}{1-\lambda}$ such that $\tilde{u}$ is in the span of the bottom t eigenvectors of $\tilde{A}$.*

Proof First, we have

$$u^\top \tilde{A} u = \frac{2}{nd}(E(\bar{I},\bar{I}) - E(I,\bar{I})) = \frac{2}{nd}(\gamma nd - (\frac{1}{2} - \gamma)nd) = -(1-4\gamma)\,.$$

Second, we derive a lower bound on $u^\top \tilde{A} u$. Let $P \in \mathbb{R}^{n\times n}$ be the orthogonal projection to the subspace spanned by the bottom t eigenvectors of $\tilde{A}$. Then, because $_{\leq -\lambda}(\tilde{A}) \leq t$ and all eigenvalues of $\tilde{A}$ lie in $[-1,1]$, we get

$$\begin{aligned} u^\top \tilde{A} u &\geq -\lambda \cdot (1 - Pu^2) + (Pu)^\top \tilde{A}(Pu) \\ &\geq -\lambda \cdot (1 - Pu^2) - Pu^2 \\ &= -\lambda - \gamma Pu^2\,. \end{aligned}$$

Then, plugging in $u^\top \tilde{A} u = -(1-4)$ and rearranging, we get

$$Pu^2 \geq \frac{1-\lambda-4\gamma}{1-\lambda} = 1 - \frac{4\gamma}{1-\lambda}\,.$$

In particular, $u - Pu^2 = 1 - Pu^2 \leq \frac{4\gamma}{1-\lambda}$. Letting $\tilde{u} = Pu$ gives the desired conclusion. $\square$

Furthermore, we can round a vector close to the indicator vector of the independent set:

Lemma 6.25 (Independent set rounding) *Let G be an n-vertex graph with vertex set V that contains an independent set $I \subseteq [n]$. Let $u \in \mathbb{R}^n$ have $u_x = \frac{1}{\sqrt{n}}$ for $x \in I$ and $u_x = -\frac{1}{\sqrt{n}}$ for $x \notin I$. Also let $\hat{u} \in \mathbb{R}^n$ satisfy $\hat{u} - u^2 \leq \alpha$ for some $\alpha \geq 0$. Then, given as input G and $\hat{u}$, there exists an algorithm with runtime (n) that returns an independent set of size at least $|I| - \alpha n$.*

Proof Letting $\hat{I} = x \in V \mid \hat{u}_x \geq 0$, by $\hat{u} - u^2 \leq \alpha$ we have $I \triangle \hat{I}(\frac{1}{\sqrt{n}})^2 \leq \alpha$. Noting that $\hat{I}$ is a superset of the independent set $\hat{I} \cap I$, by applying **??** on $\hat{I}$, we get an independent set of size at least $\hat{I} \cap I - \hat{I} \setminus (\hat{I} \cap I) = I - I \triangle \hat{I} \geq I - \alpha n$. $\square$

Given **??** and **??**, we prove **??**.

Proof (Proof of ??) We iterate over a net of the subspace spanned by the bottom t eigenvectors of $\bar{A}$ in order to find a vector close to u as defined in **??**, and then use **??** to round it to an independent set.

By standard arguments, we can construct a $c\sqrt{\gamma/(1-\lambda)}$-cover of the unit ball in the subspace spanned by the bottom t eigenvectors of $\tilde{A}$ in time $(\gamma/(1-\lambda))^{-O(t)}$, for any constant $c > 0$. The cover will have size $O(\sqrt{\gamma/(1-\lambda)})^{-t}$. Then, we iterate over all $\hat{u}$ in this cover and for each of them we apply the rounding algorithm in **??** to construct an independent set. We return the largest independent set thus obtained.

For correctness, we note by **??** that there exists some $\tilde{u}$ in the span of the bottom t eigenvectors with $\|\tilde{u}\| \leq \|u\| \leq 1$ that is $2\sqrt{\gamma/(1-\lambda)}$-close to u. Then the $c\sqrt{\gamma/(1-\lambda)}$-net that we construct is guaranteed to find some $\hat{u}$ that is $c\sqrt{\gamma/(1-\lambda)}$-close to $\tilde{u}$ and thus $(2+c)\sqrt{\gamma/(1-\lambda)}$-close to u. Then, by **??**, the returned independent set has size at least $(1/2-\gamma-(4+c')\gamma/(1-\lambda))n$. The time complexity is polynomial in n and $(\gamma/(1-\lambda))^{-O(t)}$. $\square$

6.1.6 Rounding tools

First we state a general rounding algorithm that extracts an independent set from an approximate independent set.

Lemma 6.26 (Rounding an approximate independent set) *Let G be an n-vertex graph and let $I \subseteq [n]$ be an independent set in it. Then, given a set $S \supseteq I$, there is a polynomial-time algorithm that returns an independent set I' such that $|I'| \geq |I| - |S \setminus I|$.*

Proof Note that $S \setminus I$ is a vertex cover of G_S. Hence, using the 2-approximation algorithm for the minimum vertex cover problem, we can compute a vertex cover C of G_S in polynomial time such that $|C| \leq 2|S \setminus I|$. Then we output $I' := S \setminus C$, which by construction is an independent set with size at least $|S| - 2|S \setminus I| = |I| - |S \setminus I|$, finishing the proof. $\square$

By applying **??** to k colour classes, we can extract a colouring from an approximate colouring.

Lemma 6.27 (Rounding an approximate k-colouring) *Let G be an n-vertex graph and let $\chi : [n] \to [k]$ be a 3-partition of its vertices. Suppose that G has a vertex cover of all edges with both endpoints in the same part of size at most n with respect to χ. Then, given $\chi^{-1}(1), \ldots, \chi^{-1}(k)$, there is a polynomial-time algorithm that returns a $(k, 2)$-colouring of G.*

Proof We simply run the algorithm from **??** on each set $\chi^{-1}(a)$. For each $\chi^{-1}(a)$, let I_a be the largest independent set in it. Then we obtain an independent set I'_a satisfying $|I'_a| \geq |I_a| - |\chi^{-1}(a) \setminus I_a|$. Then

$$\sum_{a=1}^{k} |I'_a| \geq \sum_{a=1}^{k} |I_a| - \sum_{a=1}^{k} |\chi^{-1}(a) \setminus I_a| \geq (1-)n - n = (1-2)n\,.$$

Chapter 7

Hardness

7.1 Hardness results

In this section, we give various hardness results against algorithms that are given as input *almost-colourable* graphs. As in **?**, our hardness reductions proceed from the following hardness result for approximating independent sets by Bansal and Khot **?**:

[Hardness of approximating independent sets **?**] Let $,\gamma > 0$ be any small enough constants. Assuming the Unique Games Conjectures, given an n-vertex graph with maximum degree $o(n)$, it is NP-hard to decide between

- the graph contains two disjoint independent sets of size $(\frac{1}{2}-)n$,
- the graph contains no independent set of size γn.

? use **??** to show that it is hard to find an independent set of size γn in a regular one-sided expander graph that admits a $(4,)$-colouring.

Our main hardness result concerns graphs colourable according to model matrices with repeated rows (see **??**). In particular, we prove that under the Unique Games Conjecture it is hard to colour all but a small fraction of the vertices of one-sided expander graphs corresponding to models with repeated rows. Our hardness result is in fact more fine-grained and matches the algorithmic result in **??**.

Theorem 7.1 (Fine-grained hardness of models) *Let $M \in \mathbb{R}^{k\times k}_{\geq 0}$ be a reversible row stochastic matrix with zero on the diagonal and non-positive second eigenvalue. Let $S_1, \dots, S_{k'}$ be the partition of $[k]$ such that, for all $a, b \in S_i$ the rows $M^{(a)}$ and $M^{(b)}$ of M are equal, and for all $a \in S_i, b \in S_j$ with $i \neq j$ the rows $M^{(a)}$ and $M^{(b)}$ of M are distinct. For all $i \in [k']$, let $a_i^* =_{a\in S_i} \pi_a$, and let $p_i = \max(0, \pi_{a_i^*} - \sum_{a\neq a_i^* \in S_i} \pi_a)$. Then, assuming the Unique Games Conjecture, the following problem is NP-hard for all $,\gamma,\lambda > 0$ small enough: Given* *a regular λ-one-sided expander graph G that admits an $(M,)$-colouring, find a $(k, 1-\sum_{i\in[k']} p_i - \gamma)$-colouring.*

As an immediate corollary, we obtain the following hardness result for colouring all but a small fraction of the vertices of one-sided expander graphs corresponding to models with repeated rows:

Corollary 7.2 (Hardness of models with repeated rows) *Let $M \in \mathbb{R}_{\geq 0}^{k\times k}$ be a reversible row stochastic matrix with zero on the diagonal and non-positive second eigenvalue, and let π be its stationary distribution. Suppose M has repeated rows. Then, assuming the Unique Games Conjecture, there exists $\delta > 0$ such that the following problem is NP-hard, for all $, \lambda > 0$: Given a regular λ-one-sided expander graph G that admits an $(M,)$-colouring, find a (k, δ)-colouring.*

We deduce two more corollaries of **??**. First, assuming the UGC, given an almost-3-colourable graph, it is NP-hard to find a proper colouring on most of its vertices. Hence, the dependence of **??** on an upper bound on the colour class sizes is necessary. Technically, this result is captured by **??** by noticing that, without an upper bound on the colour class sizes, two rows of the corresponding model matrix can be identical.

Corollary 7.3 (Hardness of unbalanced 3-colouring) *Let $, \gamma > 0$ be any small enough constants. Assuming the Unique Games Conjectures, given an n-vertex regular graph with $\lambda_2 = o(1)$ that admits a $(3,)$-colouring, it is NP-hard to find a $(3, 1/2-\gamma)$-colouring.*

The second corollary of **??** is a generalization of Proposition A.2 in **?** to any composite k. We show that, given an almost-k-colourable graph with identical colour class sizes, assuming the UGC, it is NP-hard to find a linear-sized independent set in the graph.[1]

Corollary 7.4 (Generalization of Proposition A.2 in ?) *Let $, \gamma > 0$ be any small enough constants and let k be composite. Assuming the Unique Games Conjecture, given an n-vertex regular graph with $\lambda_2 = o(1)$ that admits a $(k,)$-colouring with identical colour class sizes, it is NP-hard to find a $(k, 1-\gamma)$-colouring.*

In our final hardness result, we prove that if we only assume small λ_3, as opposed to small λ_2, then assuming the UGC, given an almost-3-colourable one-sided expander graph with identical colour class sizes, it is NP-hard to find a proper colouring on most of the vertices.

Theorem 7.5 (Hardness of balanced 3-colouring with small λ_3) *Let $, \gamma > 0$ be any small enough constants. Assuming the Unique Games Conjectures, given an n-vertex regular graph with $\lambda_3 = o(1)$ that admits a $(3,)$-colouring with identical colour class sizes, it is NP-hard to find a $(3, 2/9-\gamma)$-colouring.*

[1] It is easy to extend the lower bound to all $k \geq 4$ (not only composite k) if, instead of requiring the colour class sizes to be identical, we merely require them to be lower bounded by $\Omega(1/k)$.

We now turn to the proofs of our hardness results. To prove **??**, we first give a helper lemma showing that suitable blow-up graphs are expanding.

Lemma 7.6 (Blow-up graphs are expanding) *Let $M \in \mathbb{R}^{k\times k}_{\geq 0}$ be a reversible row stochastic matrix with zero on the diagonal and non-positive second eigenvalue. Let G be an n-vertex d-regular graph with adjacency matrix A, and let $\tilde{A} = \frac{1}{d}A$. Suppose there exists a k-partition $\chi : [n] \to [k]$ such that $M - M(\tilde{A}, \chi) = o(1)$. For $x \in [n]$ and $a \in [k]$, let $xa = \sum_{y\in\chi^{-1}(a)} \tilde{A}_{xy}$. For $a \neq b \in [k]$, let $G^{(ab)}$ be the bipartite graph induced between the vertices in $\chi^{-1}(a)$ and the vertices in $\chi^{-1}(b)$, and for $a \in [k]$ let $G^{(aa)}$ be the graph induced by the vertices in $\chi^{-1}(a)$. Suppose*

- *for all $a \in [k]$: $|\chi^{-1}(a)| \geq \Omega(n)$,*
- *for all $a, b \in [k]$: either $\lambda_1(\frac{1}{d}A_{G^{(ab)}}) = o(1)$, or $G^{(ab)}$ is biregular and $\lambda_2(\frac{1}{d}A_{G^{(ab)}}) = o(1)$.*

Then $\lambda_2(\tilde{A}) = o(1)$.

Proof For any $a, b \in [k]$, let $E^{(ab)}$ be the $n \times n$ matrix with entries $E^{(ab)}_{xy} = \tilde{A}_{xy} - \frac{M(\tilde{A},\chi)_{\chi(x)\chi(y)}}{\chi^{-1}(\chi(y))}$ for $\chi(x), \chi(y) = a, b$ and $E^{(ab)}_{xy} = 0$ otherwise. Using the definitions of the partition and model matrices in **??** and **??**, we decompose $\tilde{A}$ as

$$\tilde{A} = ZMB^{-1}Z^T + Z(M(\tilde{A}, \chi) - M)B^{-1}Z^T + \sum_{1\leq a\leq b\leq k} E^{(ab)} .$$

Then we argue about each term separately:

- the non-zero eigenvalues of M and $ZMB^{-1}Z^T$ are the same (see the proof of **??**), so because $\lambda_2(M) \leq 0$ we also have $\lambda_2(ZMB^{-1}Z^T) \leq 0$,
- because $M - M(\tilde{A}, \chi) = o(1)$ and $Z^2 = B$ we have

$$Z(M(\tilde{A}, \chi) - M)B^{-1}Z^T \leq ZM(\tilde{A}, \chi) - MB^{-1}Z^T = \frac{\max_{a\in[k]} \chi^{-1}(a)}{\min_{a\in[k]} \chi^{-1}(a)}\, o(1) = o(1) ,$$

- if for some $1 \leq a \leq b \leq k$ we have $\lambda_1(\frac{1}{d}A_{G^{(ab)}}) = o(1)$, then as $E^{(ab)}$ is obtained from $\frac{1}{d}A_{G^{(ab)}}$ by row-centering (which is an orthogonal projection) and padding with 0s, we also get $\lambda_1(E^{(ab)}) = o(1)$,
- if for some $1 \leq a \leq b \leq k$ we have that $G^{(ab)}$ is biregular and $\lambda_2(\frac{1}{d}A_{G^{(ab)}}) = o(1)$, then row-centering sends the leading eigenvalue pair to 0 while the others remain unchanged, so the spectrum of $E^{(ab)}$ is $0, 0 \cup \lambda_{2(\chi^{-1}(a)+\chi^{-1}(b)-1)}(\frac{1}{d}A_{G^{(ab)}})$, implying $\lambda_1(E^{(ab)}) = o(1)$.

Hence, as all terms are symmetric, we can use Weyl's inequality to get $\lambda_2(\tilde{A}) = o(1)$, as desired. □

We now prove **??**.

Proof (Proof of ??) For the sake of contradiction, suppose an algorithm solves the problem above efficiently and returns a $(k, 1 - \sum_{i\in[k']} p_i - \gamma)$-colouring. Let $r_1, \ldots, r_k$ be even integers bounded by $O_{M,,\gamma}(1)^2$ such that $\hat{\pi} - \pi_\infty \leq \min(\frac{\gamma}{2k}, \frac{}{2} \min_{i\in[k]} \pi_i)$ and ${}_{a\in S_i}\pi_a ={}_{a\in S_i} \hat{\pi}_a$ for all $i \in [k']$, where $\hat{\pi}_i := \frac{r_i}{\sum_{i\in[k]} r_i}$. Let $\widehat{M}$ be the $k \times k$ matrix defined by $\widehat{M}_{ij} = \frac{\hat{\pi}_j}{\pi_j} M_{ij}$, satisfying $\hat{\pi}_i \widehat{M}_{ij} = \hat{\pi}_j \widehat{M}_{ji}$. Finally, let $f : [k'] \to [k]$ be a choice function selecting an element from each set S_i, i.e., $f(i) \in S_i$ for all $i \in [k']$.

Then, we show that the following derived algorithm solves the problem in **??**:

Algorithm 3 Fine-grained model hardness algorithm

Input: A $2n$-vertex graph G with maximum degree $d_{\max} = o(n)$ from **??** with parameters $/2$ and $\frac{\gamma}{2k}$ For each $i \in [k']$ construct a graph G_i as follows: If $r_{a_i^*} \leq \sum_{a\neq a_i^* \in S_i} r_a$, let G_i have $\frac{1}{2}\sum_{a\in S_i} r_a$ disjoint copies of G Else, let G_i have $\sum_{a\neq a_i^* \in S_i} r_a$ disjoint copies of G and let it also have $(r_{a_i^*} - \sum_{a\neq a_i^*\in S_i} r_a)n$ other isolated vertices For each $1 \leq i < j \leq k'$ add a random $(\widehat{M}_{f(i)f(j)}n, \widehat{M}_{f(j)f(i)}n)$-biregular Ramanujan graph G_{ij} between $V(G_i)$ and $V(G_j)$ Decide based on whether running on the final graph $G^* = (\bigcup_{i=1}^{k'} G_i) \cup (\bigcup_{1\leq i<j\leq k'} G_{ij})$ returns a $(k, 1 - \sum_{i\in[k']} p_i - \gamma)$-colouring

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Note that by an edge removal step analogous to that in **??**, G^* can be made $\Theta(d)$-regular.

Suppose G has two disjoint independent sets of size $(1 - /2)n$ each. Then, by construction G^* admits a $(k,)$-colouring χ such that $\chi^{-1}(a) = \hat{\pi}_a V(G^*)$ for all $a \in [k]$. Indeed, if for some $i \in [k']$ we have $r_{a_i^*} \leq \sum_{a\neq a_i^*\in S_i} r_a$, then the corresponding colour classes can be mapped to independent sets in copies of G such that the two independent sets within any copy of G get a different colour. If $r_{a_i^*} > \sum_{a\neq a_i^*\in S_i} r_a$, the same holds after mapping the isolated vertices to the colour class corresponding to a_i^*.

Let $\tilde{A}^*$ be the normalized adjacency matrix of G^*. Let us show that $M(\tilde{A}^*, \chi) - M_{\max} \leq$. Let $x \in S_i$ and $y \in S_j$ be arbitrary such that $f(i) = a$ and $f(j) = b$. By the expander mixing lemma, we have $M(\tilde{A}^*, \chi)_{xy} = \widehat{M}_{ab} \pm o(1)$. On the other hand, using $\hat{\pi} - \pi_\infty \leq \frac{}{2} \min_{i\in[k]} \pi_i$ it follows that $\widehat{M}_{ab} - M_{ab} = \frac{\hat{\pi}_b}{\pi_b} M_{ab} - M_{ab} \leq \frac{}{2} M_{ab} \leq /2$. Combining the two by triangle inequality and using $M_{xy} = M_{ab}$ we get $M(\tilde{A}^*, \chi)_{xy} - M_{xy} \leq$ as desired.

Hence, in this case G^* would admit an $(M,)$-colouring. Furthermore, as the maximum degree within each G_{S_j} is $o(d)$ and the subgraphs between the G_{S_j}s are biregular Ramanujan, we can apply **??** to get $\lambda_2(G^*) = o(1)$.

[2]By **??** we have $\pi_i = \Theta(1)$ for all $i \in [k]$.

Overall we can conclude that the preconditions of would be satisfied, and by assumption it would return a $(k, 1 - \sum_{i\in[k']} p_i - \gamma)$-colouring of G^*.

Now for the contrary, assume a $(k, 1 - \sum_{i\in[k']} p_i - \gamma)$-colouring of G^*. For $j \in [k']$, let

$$q_j := \max(0, r_{a_i^*} - \sum_{a\neq a_i^* \in S_j} r_a) = \max(0, \hat{\pi}_{a_i^*} - \sum_{a\neq a_i^* \in S_j} \hat{\pi}_a) \sum_{i\in[k]} r_i \,.$$

Note that there are $\frac{(\sum_{i\in[k]} r_i - \sum_{j\in[k']} q_j)}{2}$ copies of G each of size $2n$. Then, by the pigeonhole principle over the copies of G and the colours, using $\hat{\pi} - \pi_\infty \leq \gamma/2k$, there must be an independent set of size at least

$$\begin{aligned}
&\frac{1}{k\frac{(\sum_{i\in[k]} r_i - \sum_{j\in[k']} q_j)}{2}} (\sum_{i\in[k]} r_i - \sum_{j\in[k']} q_j - (1 - \sum_{j\in[k']} p_j - \gamma) \sum_{i\in[k]} r_i) n \\
&= \frac{2n \sum_{i\in[k]} r_i}{k(\sum_{i\in[k]} r_i - \sum_{j\in[k']} q_j)} (\gamma - \sum_{j\in[k']} (\max(0, \hat{\pi}_{a_j^*} - \sum_{a\neq a_j^* \in S_j} \hat{\pi}_a) - \max(0, \pi_{a_j^*} - \sum_{a\neq a_j^* \in S_j} \pi_a))) \\
&\geq \frac{2n}{k}(\gamma - k\frac{\gamma}{2k}) \\
&= \frac{\gamma}{2k} V(G)
\end{aligned}$$

in one of the copies of G, finishing the proof. $\square$

We now provide proofs for the two corollaries of **??**.

Proof (Proof of ??) Using **??**, consider the unique model M corresponding to a stationary distribution with $\pi_1 = 1/2$, $\pi_2 = 1/4$, and $\pi_3 = 1/4$. The spectrum of M is $1, 0, -1$, so it has non-positive second eigenvalue. Noting that $\pi_2 = \pi_3$ and that the rows of M corresponding to π_2 and π_3 are identical.

Therefore, it follows from **??** that it is NP-hard under the UGC to find a $(3, 1/2 - \gamma)$-colouring of λ-one-sided expander graphs that admit an (M, ϵ)-colouring. $\square$

Proof (Proof of ??) As k is composite, it can be written as $k = k' \cdot q$ for some integers $k', q \geq 2$. Let $S_1, \ldots, S_{k'}$ be a partition of $[k]$ into groups of size q each. Let $f : [k] \to [k']$ map each $i \in [k]$ to $x \in [k']$ such that $i \in S_x$. Let us define the model matrix $M \in \mathbb{R}^{k\times k}$ with zero on the diagonal and with entries

$$M_{ij} = \begin{cases} \dfrac{1}{k-q}, & i \in S_{f(i)},\ j \notin S_{f(i)}, \\ 0, & \text{otherwise}, \end{cases}$$

for $i \neq j \in [k]$. It is easy to verify that M is row stochastic and reversible with stationary distribution $\pi = \frac{1}{k} \in \mathbb{R}^k$. The spectrum of M is 1, 0 (mult. $k - k'$), $-\frac{1}{k'-1}$ (mult. $k' - 1$) so it has non-positive second eigenvalue. By construction, for each $x \in [k']$, for all $i, j \in S_x$ the rows $M^{(i)}$ and $M^{(j)}$ of M are equal, and for all $i \in S_x, j \notin S_x$ the rows $M^{(i)}$ and $M^{(j)}$ of M are distinct.

For all $x \in [k']$, let $a_x^* :=_{a \in S_x} \pi_a$, and let $p_x := \max(0, \pi_{a_x^*} - \sum_{a \neq a_x^* \in S_x} \pi_a)$. For all $x \in [k']$, using $q \geq 2$ we have $S_x \geq 2$, so $p_x = 0$.

Therefore, as $1 - \sum_{x \in [k']} p_x - \gamma = 1 - \gamma$, it follows from **??** that it is NP-hard under the UGC to find a $(k, 1-\gamma)$-colouring of λ-one-sided expander graphs that admit an (M, ϵ)-colouring. □

We finish this section with the proof of **??**.

Proof (Proof of ??) For the sake of contradiction, suppose an algorithm solves the problem above efficiently. Then, we show that the following derived algorithm solves the problem in **??**:

Algorithm 4 Low top-threshold rank hardness algorithm

Input: The n-vertex graph G with average degree $d = o(n)$ and maximum degree $d_{\max} = o(n)$ from **??** with parameters $, \frac{3}{2}\gamma$ Insert into the graph $\frac{n+\sqrt{n(n-4d)}}{2}$ new vertices and denote their set by S Add edges H between $V(G)$ and S to form a complete bipartite graph Let H' be a bipartite graph between $V(G)$ and S with degrees $d_G(x)$ for $x \in V(G)$ and $\frac{nd}{S}$ for $y \in S$ Remove H' from H to obtain a graph G' Let G'' be a new S-regular tripartite graph with $\lambda_2(A_{G''}) = o(S)$ and sides of size $S, S, (\frac{1}{2} -)n$ [3] Decide based on whether running on the final graph $G^* = G' \cup G''$ returns a $(3, \frac{2}{9} - \gamma)$-colouring of G^*

Let us first show that G^* is S-regular. Note that the degrees of $x \in V(G)$ coming from G are offset by the removal of H', so they will exactly have degree S coming from H. For a vertex $y \in S$, the degree coming from H is n, and the degree removed by H' is $\frac{nd}{S}$, so the degree of y in G^* is also

$$n - \frac{nd}{S} = n - \frac{nd}{\frac{n+\sqrt{n(n-4d)}}{2}} = \frac{n + \sqrt{n(n-4d)}}{2} = S\,.$$

[3] Using **??** and $S \geq n/4$, such a graph exists based on the model matrix $\begin{pmatrix} 0 & S-(1/4-/2)n & (1/4-/2)n \\ S-(1/4-/2)n & 0 & (1/4-/2)n \\ åS/2 & S/2 & 0 \end{pmatrix}$ with a non-positive second eigenvalue.

Decomposing $A_{G'} = A_G + A_H - A_{H'}$ and using that $\lambda_2(A_H) = 0$ and that G and H' both have maximum degree at most $d_{\max} = o(n)$, we get by Weyl's inequality that

$$\lambda_2(A_{G'}) \leq \lambda_1(A_G) + \lambda_2(A_H) - \lambda_n(A_{H'}) = o(n)\,.$$

Using $d = o(n)$ we have $|S| = \Omega(n)$, so $\lambda_2(A_{G'}) = o(|S|)$. Then, decomposing $A_{G^*} = A_{G'} + A_{G''}$ and using that $\lambda_2(A_{G''}) = o(|S|)$ by construction, we finally get by Weyl's inequality that

$$\lambda_3(A_{G^*}) \leq \lambda_2(A_{G'}) + \lambda_2(A_{G''}) = o(|S|)\,.$$

If G has two disjoint independent sets of size $(\frac{1}{2}-)n$, then G^* admits a with identical colour class sizes on all but $2n \leq |V(G^*)|$ vertices. Hence, using the $|S|$-regularity of both G' and G'' and that $\lambda_3(G^*) = o(1)$, returns a $(3, \frac{2}{9} - \gamma)$-colouring of G^*.

However, we show that if G^* admits a $(3, \frac{2}{9} - \gamma)$-colouring of G^*, then G has an independent set of size $\frac{3}{2}\gamma n$, and is a distinguisher. Indeed, note that a $(3, \frac{2}{9} - \gamma)$-colouring of G^* implies an independent set in G of size at least

$$\frac{1}{3}\left((\frac{7}{9} + \gamma)V(G^*) - (S + |V(G'')|) \right),$$

which after plugging in the size of the components, and using that $S \leq n$, is lower bounded by $\frac{3}{2}\gamma n$. □

Bibliography

Noga Alon and Nabil Kahale. A spectral technique for coloring random 3-colorable graphs. *SIAM J. Comput.*, 26(6):1733–1748, 1997.

Mitali Bafna, Jun-Ting Hsieh, and Pravesh K Kothari. Rounding large independent sets on expanders. *arXiv preprint arXiv:2405.10238*, 2024.

Nikhil Bansal and Subhash Khot. Optimal long code test with one free bit. In *2009 50th Annual IEEE Symposium on Foundations of Computer Science—FOCS 2009*, pages 453–462. IEEE Computer Soc., Los Alamitos, CA, 2009.

Nikhil Bansal and Subhash Khot. Optimal long code test with one free bit. In *2009 50th Annual IEEE Symposium on Foundations of Computer Science—FOCS 2009*, pages 453–462. IEEE Computer Soc., Los Alamitos, CA, 2009.

Boaz Barak, Prasad Raghavendra, and David Steurer. Rounding semidefinite programming hierarchies via global correlation. In *2011 IEEE 52nd Annual Symposium on Foundations of Computer Science—FOCS 2011*, pages 472–481. IEEE Computer Soc., Los Alamitos, CA, 2011.

Avrim Blum and Joel Spencer. Coloring random and semi-random k-colorable graphs. *J. Algorithms*, 19(2):204–234, 1995.

Roee David and Uriel Feige. On the effect of randomness on planted 3-coloring models. In *STOC'16—Proceedings of the 48th Annual ACM SIGACT Symposium on Theory of Computing*, pages 77–90. ACM, New York, 2016.

Richard Karp. Reducibility among combinatorial problems (1972). In *Ideas that created the future—classic papers of computer science*, pages 349–356. MIT Press, Cambridge, MA, [2021] ©2021. Reprinted from [0378476].

Ken-ichi Kawarabayashi, Mikkel Thorup, and Hirotaka Yoneda. Better coloring of 3-colorable graphs. In *STOC'24—Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pages 331–339. ACM, New York, [2024] ©2024.

Akash Kumar, Anand Louis, and Madhur Tulsiani. Finding pseudorandom colorings of pseudorandom graphs. In *37th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2017)*, pages 37–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2018.

Martin J. Wainwright. *High-dimensional statistics*, volume 48 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge, 2019. A non-asymptotic viewpoint.

Proof of top threshold rank lower bound

In this section, we prove for the sake of completeness a slight generalization of Lemma 6.1 in **?**, such that it extends to all bounded-norm symmetric matrices. The proof is essentially the same as the original proof. We note that the original proof has a minor mistake in an application of Cauchy-Schwarz, and to fix it we need to modify the guarantee of the lemma to give a threshold rank lower bound of $(1-1/C)^2 r$ instead of the original lower bound of $(1-1/C)r$.

Lemma .7 (Generalized restatement of Lemma 6.1 in ?) *Let $A \in \mathbb{R}^{n\times n}$ be symmetric with $A \leq 1$. Suppose there exists $M \in \mathbb{R}^{n\times n}$ positive semidefinite such that*

$$A, M \geq 1- \,, \quad M_F^2 \leq \frac{1}{r}\,, \quad (M) = 1\,.$$

Then for all $C > 1$,

$${}_{\geq 1-C\cdot}(A) \geq (1-\frac{1}{C})^2 r\,.$$

Proof Let the spectral decomposition of A be $A = \sum_i \lambda_i u_i u_i^\top$ and let $k =_{\geq 1-C}$ (A). We use $\Lambda_{\geq 1-C} = \sum_{i=1}^k u_i u_i^\top$ to denote the subspace spanned by the eigenvectors with corresponding eigenvalues at least $1 - C$. Then we can rewrite A, M as

$$\begin{aligned} A, M &= \sum_{i=1}^{k} \lambda_i u_i u_i^\top, M + \sum_{i=k+1}^{n} \lambda_i u_i u_i^\top, M \\ &\leq \Lambda_{\geq 1-C}, M + (1-C)_n - \Lambda_{\geq 1-C}, M \\ &= C\Lambda_{\geq 1-C}, M + (1-C)_n, M\,. \end{aligned} \tag{.1}$$

For the first term, it follows by the assumption $M_F^2 \leq \frac{1}{r}$ and Cauchy–Schwarz that

$$\Lambda_{\geq 1-C}, M \leq \Lambda_{\geq 1-C\,F} M_F \leq \sqrt{k}\cdot\frac{1}{\sqrt{r}}\,. \tag{.2}$$

For the second term, it follows by the assumption $(M) = 1$ that

$${}_n, M = (M) = 1\,. \tag{.3}$$

Plugging **??** and **??** into **??**, it follows that

$$A, M \leq C\sqrt{\frac{k}{r}} + 1 - C\,.$$

Since by assumption $A, M \geq 1-$, it follows by rearranging the terms that

$$_{\geq 1-C\cdot}(A) = k \geq (1 - \frac{1}{C})^2 r\,.$$

Full colouring in the random planted model

This part is dedicated to proving **??**. We start with the key first step that guarantees the recovery of a *list* of candidate k-colourings, such that at least one of them is close to the planted k-colouring.

Theorem .8 (Result for random planting, partial recovery of a list) *Let H be an n-vertex d-regular graph with adjacency matrix A_H. Suppose $_{\geq\lambda}(\frac{1}{d}A_H) \leq t$ for some $0 < \lambda < 1$ and $t \in \mathbb{N}$. Let G be the graph obtained by randomly planting a in H, and let $\chi : [n] \to [k]$ be the associated . Then, for $\lambda > 0$ small enough, there exists an algorithm that runs in time $n^{O(1)} \cdot (O_k(1))^t$ and outputs a list of $(O_k(1))^t$ k-colourings $\hat{\chi} : [n] \to [k]$ such that with high probability, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim\mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O_k(1/d^{\Omega(1)})$.*

Proof The proof is identical to the proof of **??**, but omitting the last rounding step. □

To convert a colouring returned by **??** into a full colouring, we will need several helper lemmata. We start with the proofs of the two lemmata showing properties of **one**-sided expansion.

Proof (Proof of ??) Let $_S \in \{0,1\}^n$ be the indicator vector of S and write $_S$ in the orthonormal eigenbasis $e_{[n]}$ of A:

$$_S = \sum_{i=1}^{n} \alpha_i e_i\,.$$

Hence, by orthonormality, we can write

$$\begin{aligned}
&E(H_S)\\
&= \frac{1}{2}_S^{\top} A_S\\
&= \frac{1}{2}(\sum_{i=1}^{n} \alpha_i e_i^{\top})(\sum_{i=1}^{n} \lambda_i(A) e_i e_i^{\top})(\sum_{i=1}^{n} \alpha_i e_i)\\
&= \frac{1}{2}\sum_{i=1}^{n} \lambda_i(A)\alpha_i^2
\end{aligned}$$

Using d-regularity, we have $\lambda_1(A) = d$. Then, as $e_1 = \frac{1}{\sqrt{n}}$, we get $\alpha_1 = e_{1},_S = \frac{|S|}{\sqrt{n}}$, so $\lambda_1(A)\alpha_1^2 = \frac{d|S|^2}{n}$. For the terms with $i > 1$, as $\alpha_i^2 \geq 0$, we can upper bound $\lambda_i(A)\alpha_i^2$ by $\lambda_2(A)\alpha_i^2 \leq cd\alpha_i^2$. Plugging these back we get

$$E(H_S) \leq \frac{1}{2}(\frac{d|S|^2}{n} + cd\sum_{i=2}^{n} \alpha_i^2)\,.$$

Then, using $\sum_{i=2}^{n} \alpha_i^2 \leq \sum_{i=1}^{n} \alpha_i^2 = s_2^2 = |S|$ gives the desired bound. $\square$

Proof (Proof of ??) Let $T := N(S) \cup S$. Then, as all edges touching S are in $E(H_T)$, we have

$$e(H_T) \geq \frac{1}{2}|S|d\,.$$

On the other hand, by **??**, writing $|T| = x|S|$ and using $|S| \leq \alpha n$, we get

$$e(H_T) \leq \frac{d|T|}{2}(\frac{|T|}{n} + c) \leq \frac{1}{2}x(c + \alpha x)|S|d\,.$$

Comparing the two bounds, and solving the resulting quadratic inequality, we get

$$x \geq \frac{\sqrt{c^2 + 4\alpha} - c}{2\alpha} = \frac{2}{c + \sqrt{c^2 + 4\alpha}}\,.$$

By using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and that $N(S) \setminus S \geq N(S) - |S|$ we get the desired result. $\square$

Next, we need an algorithm that takes a colouring that agrees with most of the planted colouring and turns it into a colouring that leaves some of the vertices uncoloured, but agrees with the planted colouring on the coloured vertices. We define this notion of colouring formally now.

Definition .9 (Partial colouring) *Let G be an n-vertex graph that admits a χ. A (k,β)-*partial *colouring w.r.t. χ is a of at least $(1-\beta)n$ vertices that is identical to χ on these vertices. The remaining βn vertices are left uncoloured and are referred to as* free. *In the random planted model we always implicitly take χ to be the planted .*

Let us state now the corresponding algorithm and its guarantees.

Algorithm 5 Uncolouring Algorithm

As long as there is a vertex x that has colour c_1, and a colour $c_2 \neq c_1$ such that x has less than $\frac{1}{6k}d$ neighbors with colour c_2, uncolour x.

Lemma .10 (Adaptation of Lemma B.20 of ?) *Let $\alpha \leq \frac{1}{24k}$ be a parameter. Let H be a $d \geq O(1)$-regular n-vertex graph with $\lambda_2(H) \leq \frac{1}{9k}$. Let G be the graph obtained from H by randomly planting a χ in it. Given an $\hat{\chi}$ of G with $\mathbb{E}_{x\sim\mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq \alpha$,* **??** *returns an $(k, 4\alpha)$-partial colouring of G.*

To prove **??**, we will need a definition and two lemmata: one upper bounding the number of atypical vertices with high probability, and a structural lemma showing the existence of a "nice" subgraph.

Definition .11 *Let H be a d-regular graph. Let G be the graph obtained from H by randomly planting a χ in it. We call a vertex $x \in V(G)$ statistically bad, if there is a colour $c \neq \chi(x)$ such that x has less than $\frac{1}{2k}d$ neighbors y with $\chi(y) = c$. We denote the set of statistically bad vertices by $SB(G)$, writing SB when it is clear from context which graph we are referring to.*

Lemma .12 (Restatement of Lemma B.14 of ?) *Let H be a $d \geq O(1)$-regular n-vertex graph. Let G be the graph obtained from H by randomly planting a in it. It holds that $SB(G) \leq n2^{-\Omega(d)}$.*

Lemma .13 (Adaptation of Lemma B.19 of ?) *Let $0 < \alpha, c < 1$ be parameters. Let H be a d-regular n-vertex graph with $\lambda_2(H) \leq c$. For any $S \subseteq V$ with $S < \alpha n$, there exists $R \subseteq V \setminus S$ such that $R \geq (1-2\alpha)n$ and the minimum degree in H_R is at least $(1-2\alpha-c)d$.*

Proof Let $:= 2\alpha + c$. Consider the following iterative procedure. Repeatedly remove a vertex v_t from $V \setminus (S \cup \bigcup_{i=1}^{t-1} v_i)$ if v_t has more than d neighbors in $S_t := S \cup \bigcup_{i=1}^{t-1} v_i$. Assume towards a contradiction that this process stops after more than $t := S$ steps. Then, by using the vertices in $S_t \setminus S$, we can lower bound the number of edges in H_{S_t}:

$$e(H_{S_t}) \geq S_t \setminus Sd = \frac{1}{2}S_t d\,.$$

On the other hand, be **??**,

$$e(H_{S_t}) \leq \frac{S_t d}{2}(\frac{S_t}{n} + c)\,.$$

Comparing the two and using $S = \frac{1}{2}S_t$ we get

$$S \geq \frac{1}{2}(-c)n = \alpha n$$

which contradicts upper bound assumption on S. Hence, letting $R := V \setminus S_t$ it must hold that $R \geq n - 2S \geq (1-2\alpha)n$ and the minimum degree in H_R is at least $(1-)d = (1-2\alpha-c)d$ by construction. $\square$

Now we are ready to prove **??**.

Proof (Proof of ??) Let $c := \frac{1}{9k}$. When d is at least a large enough constant, $2^{-\Omega(d)} \leq \alpha$ holds. Let $B := x \in V \mid \chi(x) \neq \hat{\chi}(x)$ denote the set of vertices wrongly coloured by $\hat{\chi}$. Note that $B \leq \alpha n$ by assumption, and $SB(G) \leq 2^{-\Omega(d)} \leq \alpha n$ by **??**. Hence, applying **??** on H to $B \cup SB$ we get a set R with $|R| = (1-4\alpha)n$ such that the minimum degree in H_R is at least $(1-4\alpha-c)d$. Using $R \cap SB = \emptyset$, for any $x \in R$ and $c \neq \chi(x)$ we get that x has degree at least $(\frac{1}{2k} - 4\alpha - c)d \geq \frac{1}{6k}d$ into $y \in R \mid \chi(y) = c$, i.e. into the vertices in R coloured with c. It follows by induction that no vertices in R will be uncoloured at any step of **??**. Hence, the colouring returned has size at least $|R| = (1-4\alpha)n$. It only remains to show that the colouring returned is proper. In fact, we will show that all vertices in B will get uncoloured, from which the statement follows.

For the sake of contradiction, assume there exist vertices $B_2 \subseteq B$ that remain coloured by the end of the process. Let $x \in B_2$ be any such vertex. As x has never been uncoloured, at the end of the process it has at least $\frac{1}{6k}d$ neighbors y with $\hat{\chi}(y) = c \neq \hat{\chi}(x)$. In particular, choosing $c = \chi(x)$, let Q denote those neighbors y of x with $\hat{\chi}(y) = \chi(x)$. As x and y are neighbors in G, we have $\chi(x) \neq \chi(y)$, so $\hat{\chi}(y) \neq \chi(y)$, which implies $Q \subseteq B_2$. Summarizing, any vertex in G_{B_2} has degree at least $\frac{1}{3k}d$. Therefore,

$$e(G_{B_2}) \geq \frac{\frac{1}{6k}d|B_2|}{2} = \frac{d|B_2|}{12k}\,.$$

On the other hand, by **??**,

$$e(G_{B_2}) \leq e(H_{B_2}) \leq \frac{dB_2}{2}(\frac{B_2}{n} + c)\,.$$

Using $B_2 > 0$ and the bounds on α and c, we can combine and simplify the above two inequalities to get

$$B_2 \geq (\frac{1}{6k} - c)n > \alpha n\,.$$

However, as $B_2 \leq B \leq \alpha n$ by assumption, we get a contradiction, so $B_2 = 0$ finishing the proof. □

Next we need an algorithm that recolours some of the uncoloured vertices in a way that guarantees that the remaining uncoloured vertices split up into small components.

Algorithm 6 Safe Recolouring Algorithm

As long as there is an uncoloured vertex x that has neighbors in every colour except for some colour c, colour x using c.

Lemma .14 (Adaptation of Lemma B.21 of ?) *Let H be a d-regular n-vertex graph with $\lambda_2(H) \leq \frac{1}{16k^2}$. Let G be the graph obtained from H by randomly planting a χ in it. Starting from a $(k, \frac{1}{256k^4})$-partial colouring $\hat{\chi}$ of G, let U be the set of vertices left uncoloured by* **??**. *Then, each component in G_U has size at most $\ln n$.*

Proof The proof closely follows that of Lemma B.21 of **?**, so we only provide the parts that are changed. Let us first prove a helper lemma similar to Claim B.22 of **?**:

Lemma .15 *Let C be any set of vertices of size t_1, let $D := N_H(C) \setminus C$ and let $t_2 := D$. If $t_2 \geq 2k^2 t_1$ then $C \subseteq U$ and $D \subseteq V \setminus U$ with probability at most $e^{-\frac{1}{2k}t_2}$.*

Proof Partition the vertices of D to disjoint subsets D_v, where $v \in C$ and $D_v \subseteq N_H(v)$. Consider some D_v and assume that v is coloured by 1. If both $v \in U$ and $D_v \subseteq V \setminus U$ then the set of colours that vertices in D_v are using must be contained in exactly one of the sets $[k] \setminus c$ for some $c \neq 1$ (otherwise **??** would colour v). Therefore the probability that both $v \in U$ and $D_v \subseteq V \setminus U$ is at most $(k-1)(\frac{k-1}{k})^{D_v}$. Hence, the probability that both C is in U and $D \subseteq V \setminus U$ is at most

$$\begin{aligned}
\prod_{D_v \in D_v} (k-1)(\frac{k-1}{k})^{D_v} &= (k-1)^{D_v}(\frac{k-1}{k})^{t_2} \\
&\leq (k-1)^{t_1}(\frac{k-1}{k})^{t_2} \\
&= e^{(t_1+t_2)\ln(k-1) - t_2 \ln k} \\
&\leq e^{-\frac{1}{2k}t_2}
\end{aligned}$$

using $t_2 \geq 2k^2 t_1$ and $\ln k - \ln(k-1) \geq \frac{1}{k}$ on the last line. □

Letting $N_H(n, t_1, t_2)$ denote the number of connected components of size t_1 with a neighborhood of size exactly t_2 and letting $P_{t_1,t_2} := \max_{C \subset H, C = t_1, N(C) = t_2} \Pr C \text{ is in } U$, we get by union bound that the probability that there exists a connected component in G_U of size $\Omega(\ln n)$ is at most

$$\sum_{t_1=\ln n}^{U}\sum_{t_2=1}^{dt_1} N_H(n,t_1,t_2)P_{t_1,t_2} = \sum_{t_1=\ln n}^{U}\sum_{t_2=7k^2t_1}^{dt_1} N_G(n,t_1,t_2)e^{-\frac{1}{2k}t_2} \tag{.4}$$

$$\leq \sum_{t_1=\ln n}^{U}\sum_{t_2=7k^2t_1}^{dt_1} n\binom{t_1+t_2}{t_1}e^{-\frac{1}{2k}t_2} \tag{.5}$$

$$\leq \sum_{t_1=\ln n}^{U}\sum_{t_2=7k^2t_1}^{dt_1} n(e\frac{t_1+t_2}{t_1})^{t_1}e^{-\frac{1}{2k}t_2} \tag{.6}$$

$$\leq \sum_{t_1=\ln n}^{U} ndt_1 \max_{7k^2\leq x\leq d} e^{(1+\ln(x+1)-\frac{1}{2k}x)t_1}$$

$$\leq n^4 \max_{7k^2\leq x\leq d} e^{(1+\ln(x+1)-\frac{1}{2k}x)\ln n}$$

$$\leq n^{-1}. \tag{.7}$$

Equality **??** follows by **??** and that by **??** if $\lambda_2(H) \leq \frac{1}{16k^2}$ then for a set of size $t_1 \leq U \leq \frac{1}{256k^4}n$ it holds that its neighborhood is of size $t_2 \geq 7k^2t_1$. Inequality **??** follows by Lemma B.23 of **?**. Inequality **??** follows by the binomial identity $\binom{n}{k} \leq (e\frac{n}{k})^k$. Inequality **??** follows by $\ln(7k^2+1) \leq \frac{7k}{2} - 6$ which holds for $k \geq 3$. This completes the proof. $\square$

We are now finally ready to prove **??**:

Proof (Proof of ??) We can apply **??** to get a list of $O_k(1)$ k-colourings $\hat{\chi}$: $[n] \to [k]$ such that, for at least one of them, it holds up to a permutation of the colour classes that $\mathbb{E}_{x\sim\mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O_k(1/d^{\Omega(1)})$.

Then, for d at least a large enough constant compared to k, running **??** on all colourings in L, by **??** it turns the colouring $\hat{\chi}$ with $\mathbb{E}_{x\sim\mathrm{Unif}([n])}[\hat{\chi}(x) \neq \chi(x)] \leq O_k(1/d^{\Omega(1)})$ into a $(k, O_k(1/d^{\Omega(1)}))$-partial colouring and returns a list of colourings L'.

Then, for d at least a large enough constant compared to k, running **??** on all colourings in L', **??** guarantees that, for the $(k, O_k(1/d^{\Omega(1)}))$-partial colouring, all the remaining still uncoloured vertices are in connected components of size $O(\ln n)$ and returns a list of colourings L'':

Hence, we can try a brute force search on each component of those colourings in L'' that have component sizes $O(\ln n)$. By the above, we are guaranteed to find a full colouring on at least one of them. This finishes the proof. $\square$

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